

Network Theorem

Star-Delta conversion

Superposition theorem

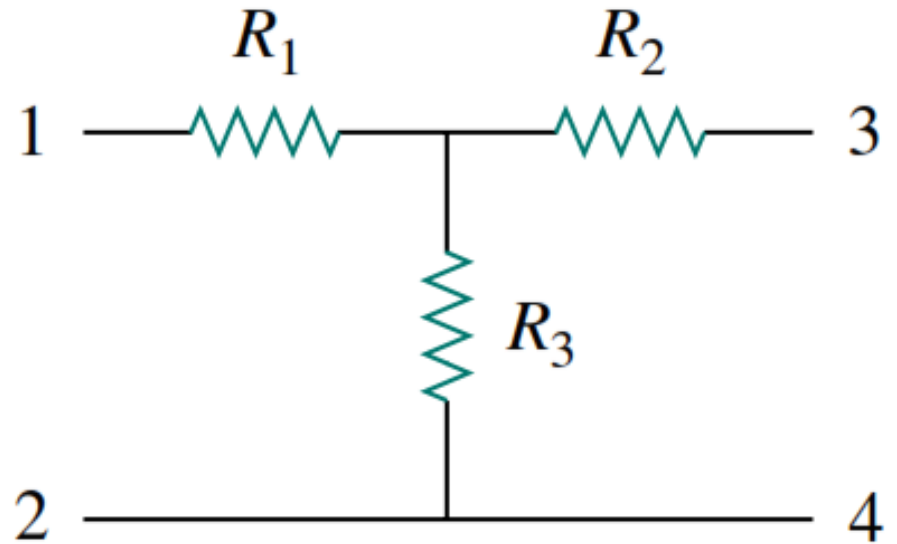
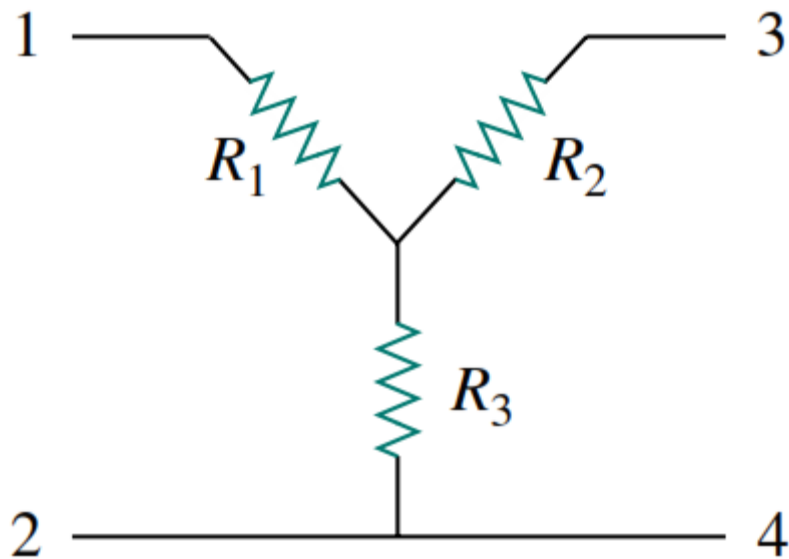
Thevenin theorem

Maximum Power Transfer Theorem

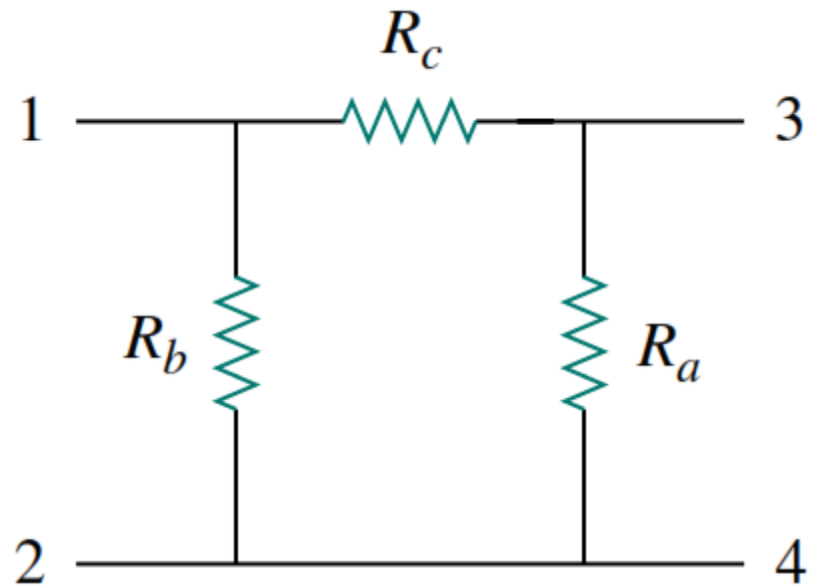
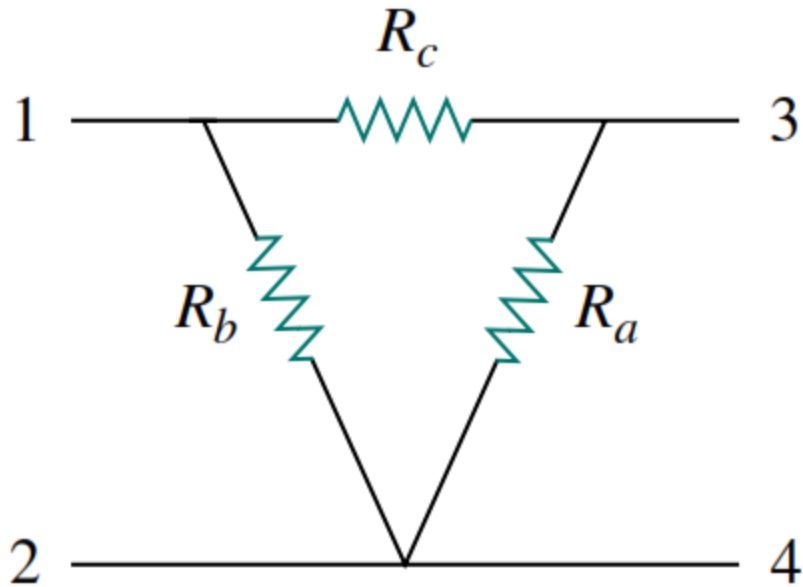
Our Lord! Grant us good in this world and good in the hereafter, and save us from the chastisement of the fire. [2:201]

Star-Delta conversion

Y or T



Δ or Π

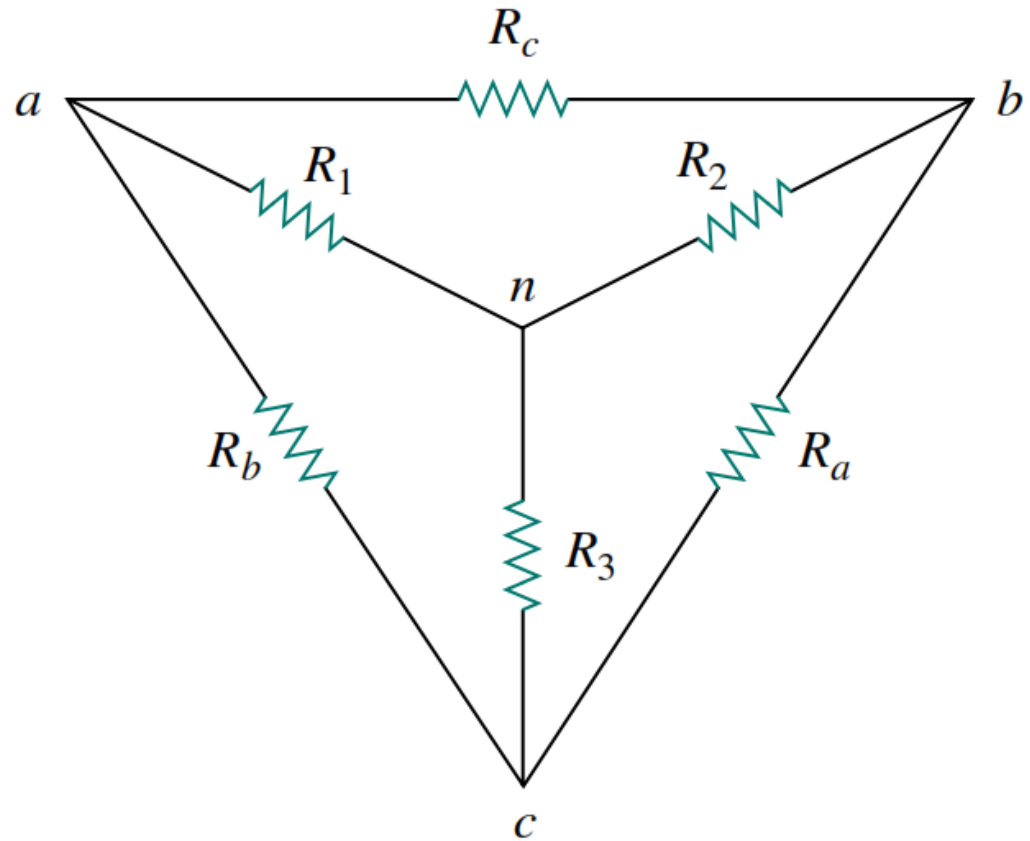


Δ to Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$



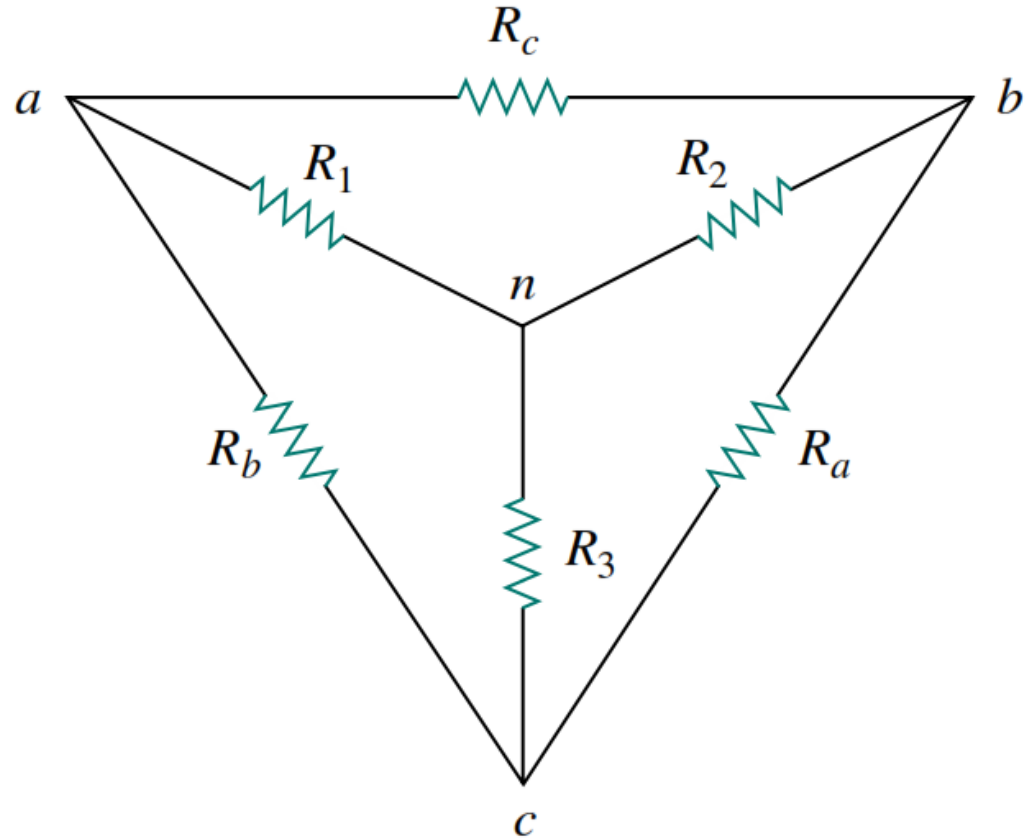
Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

Y to Δ

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

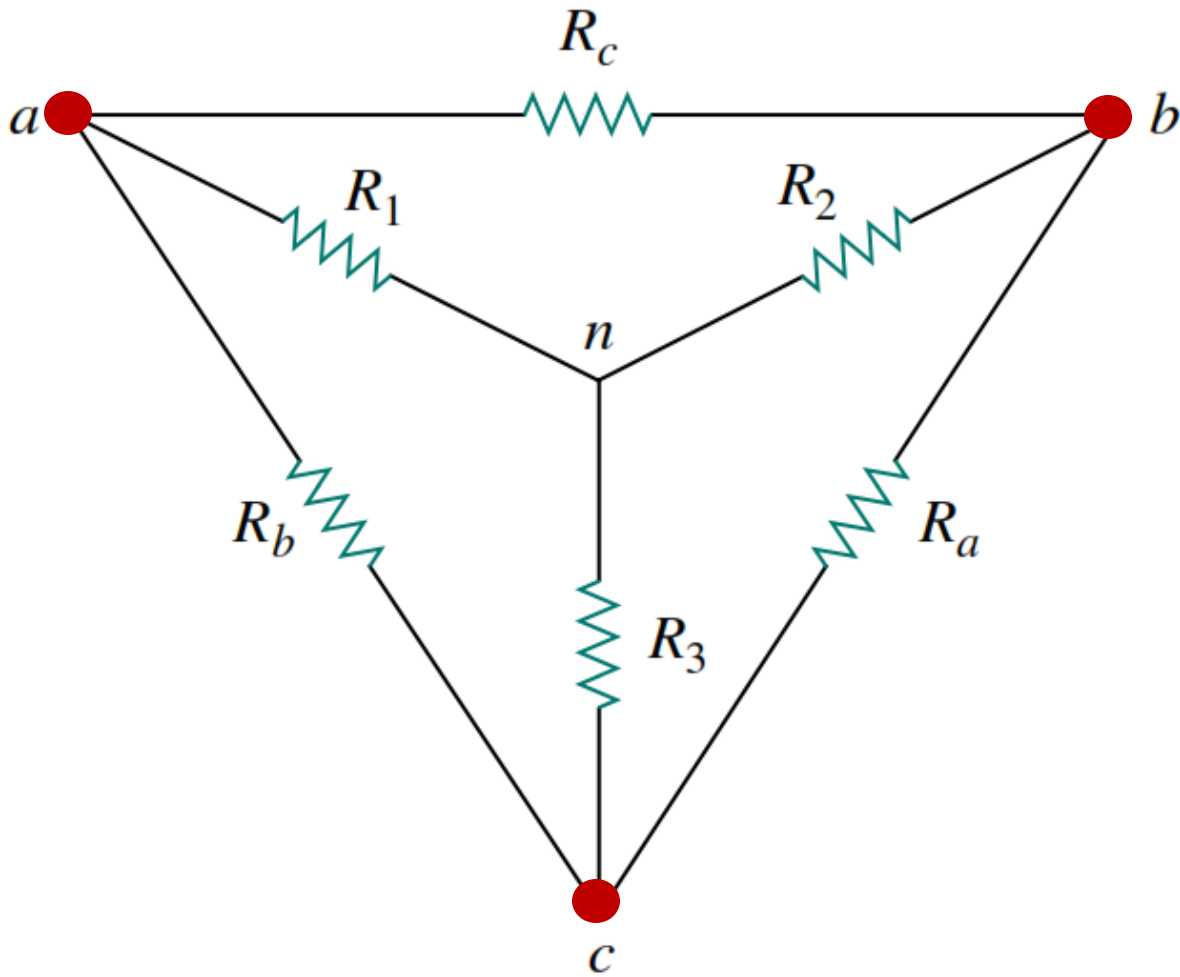
$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

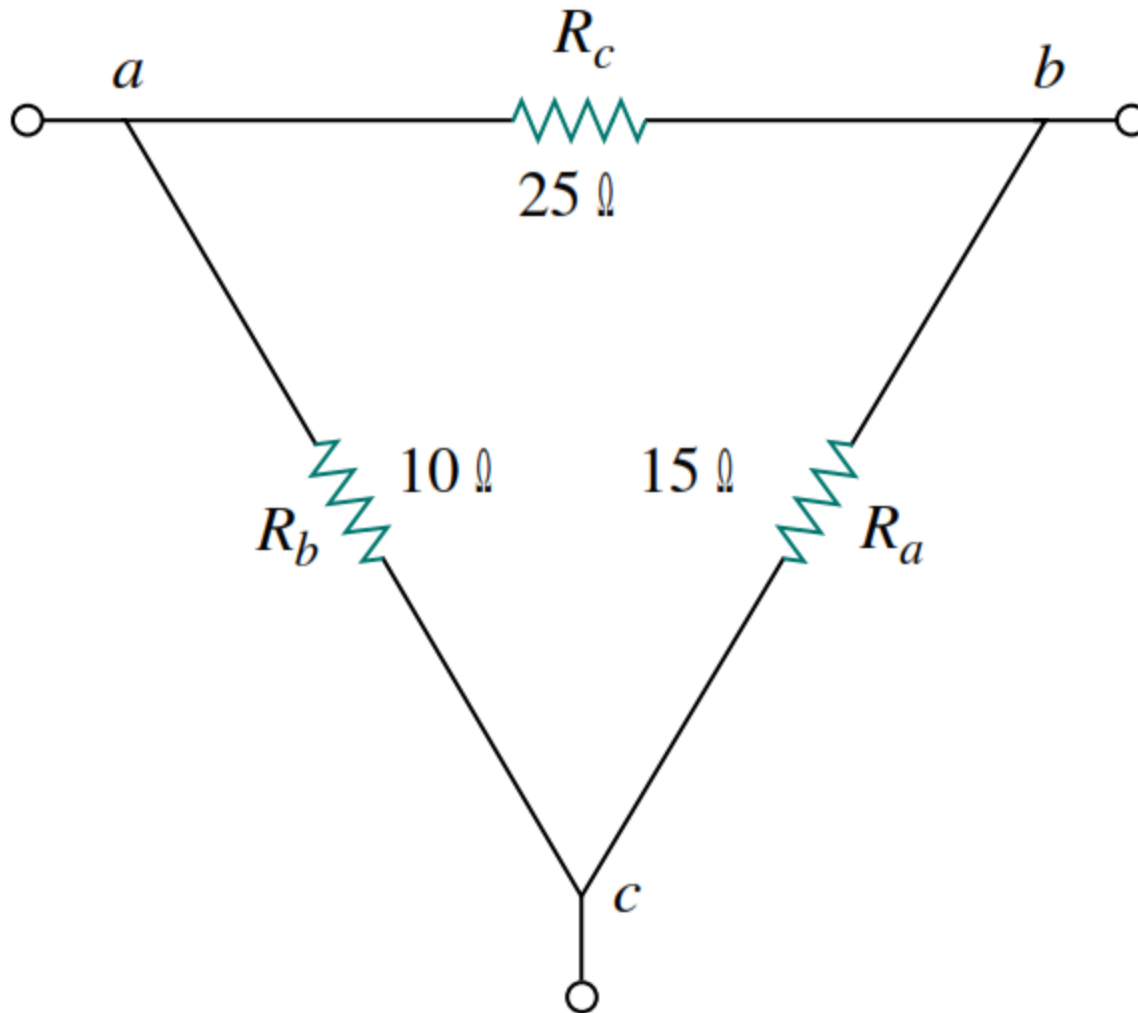


Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

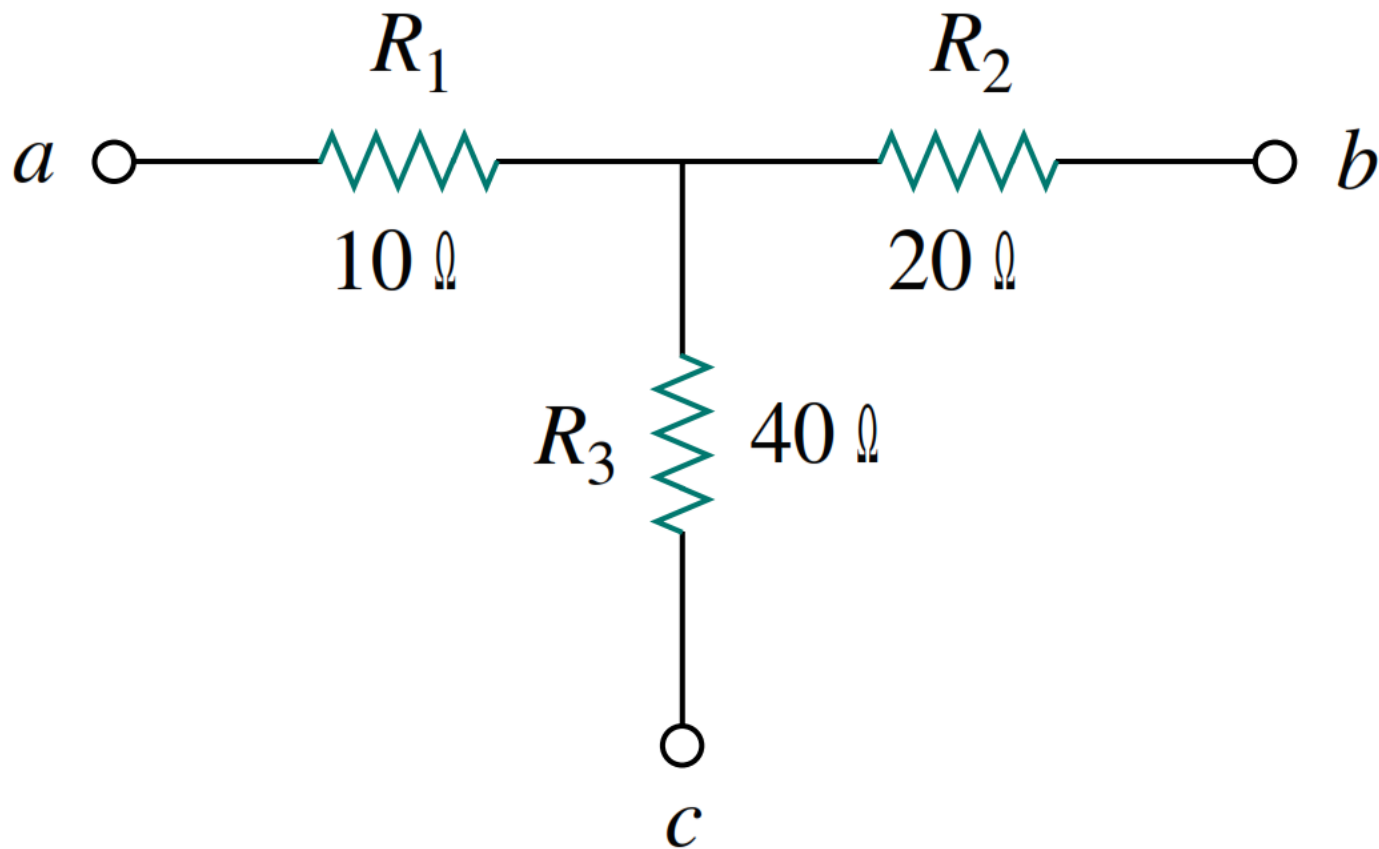
Remember



$$Y = ?$$

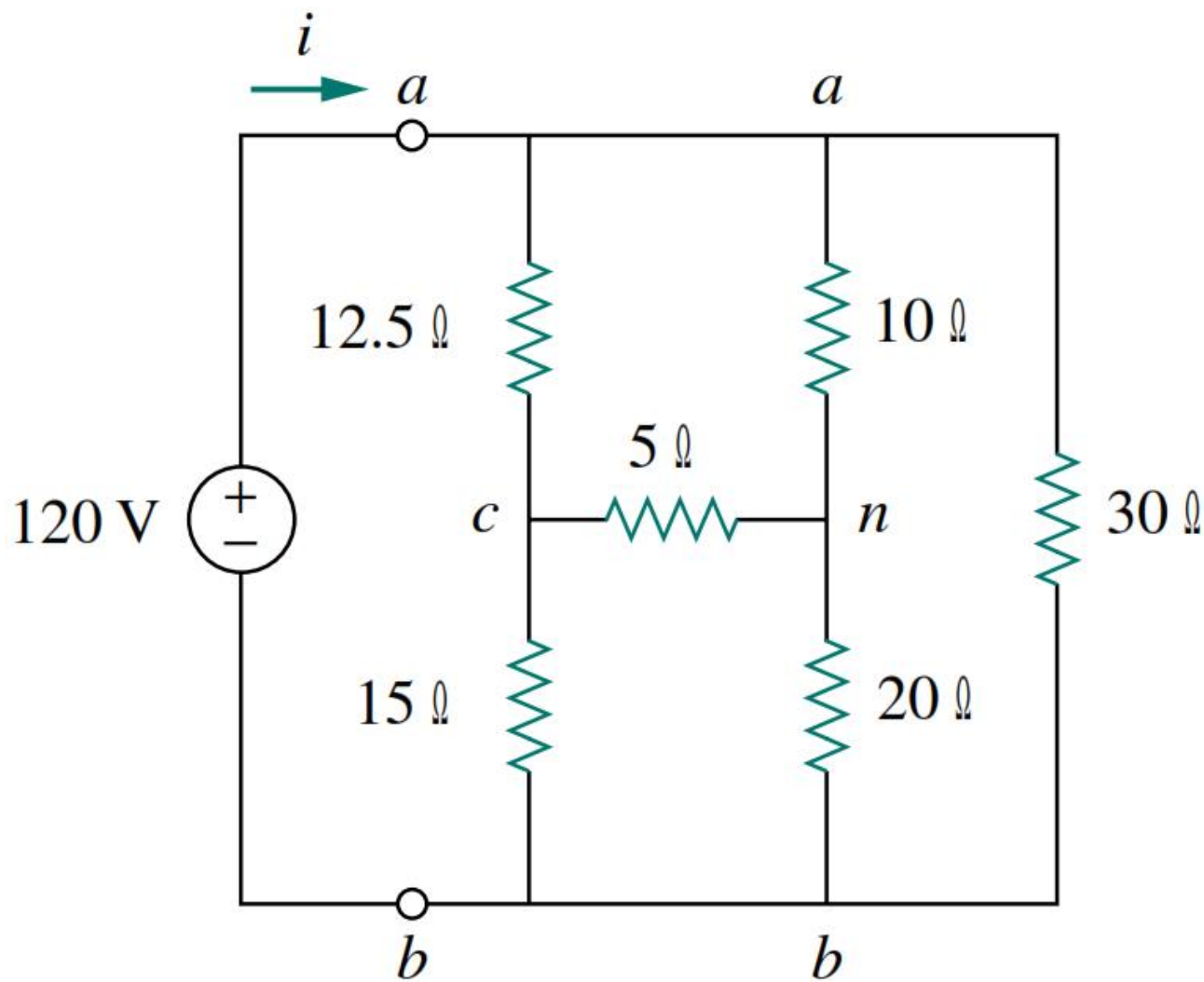


$$\Delta = ?$$



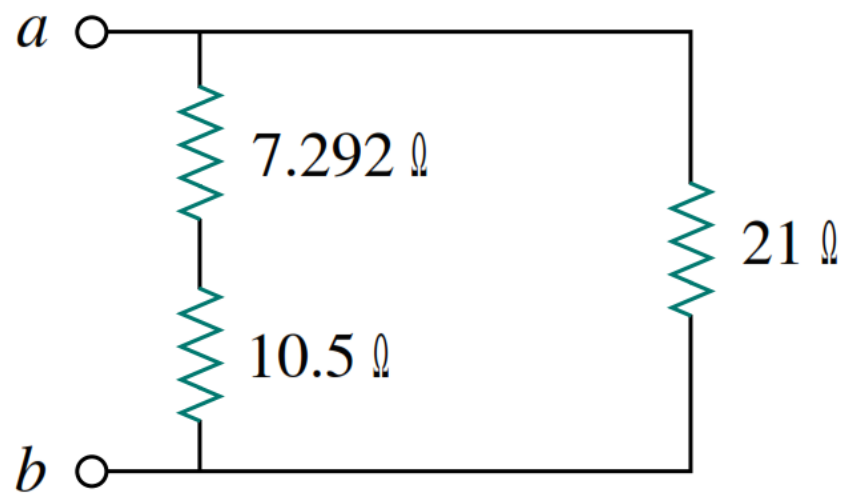
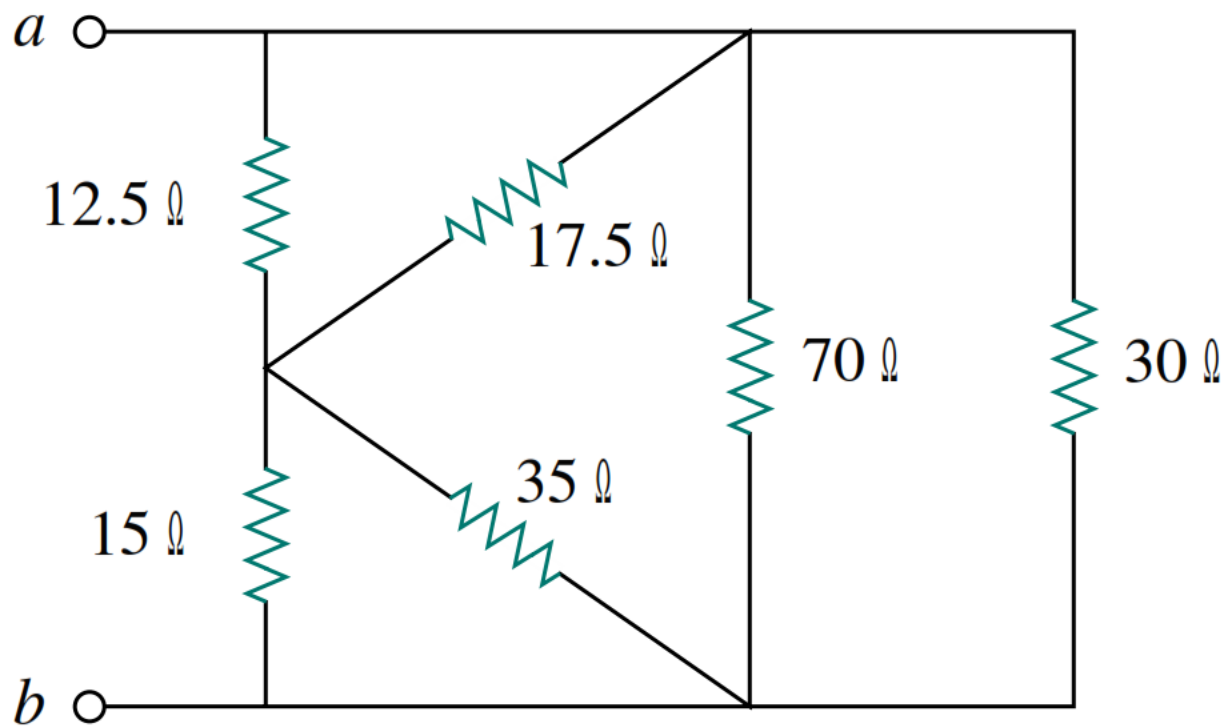
$$R_{ab} = ?$$

$$i = ?$$



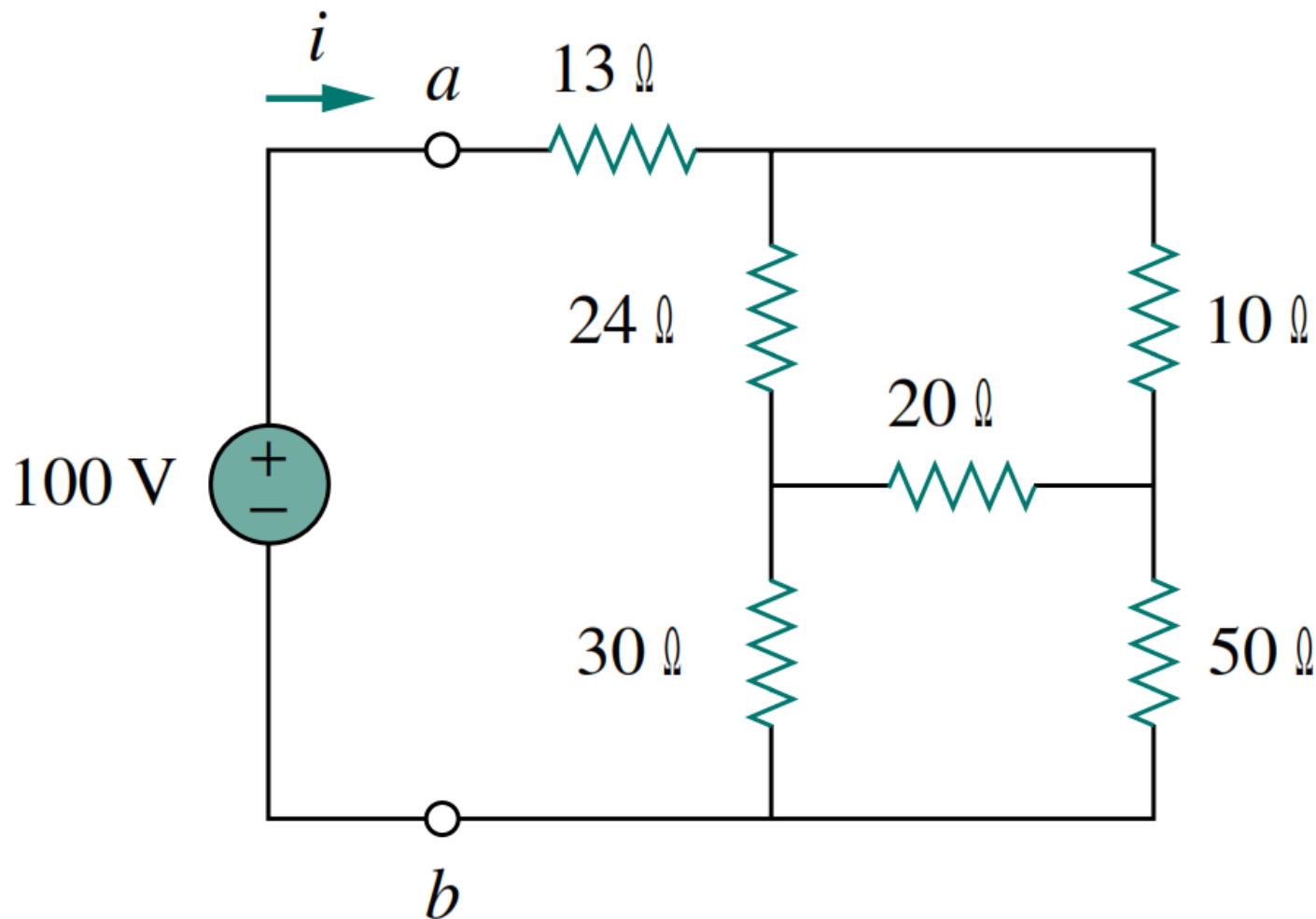
$$9.632 \, \Omega$$

$$12.458 \, \text{A}$$

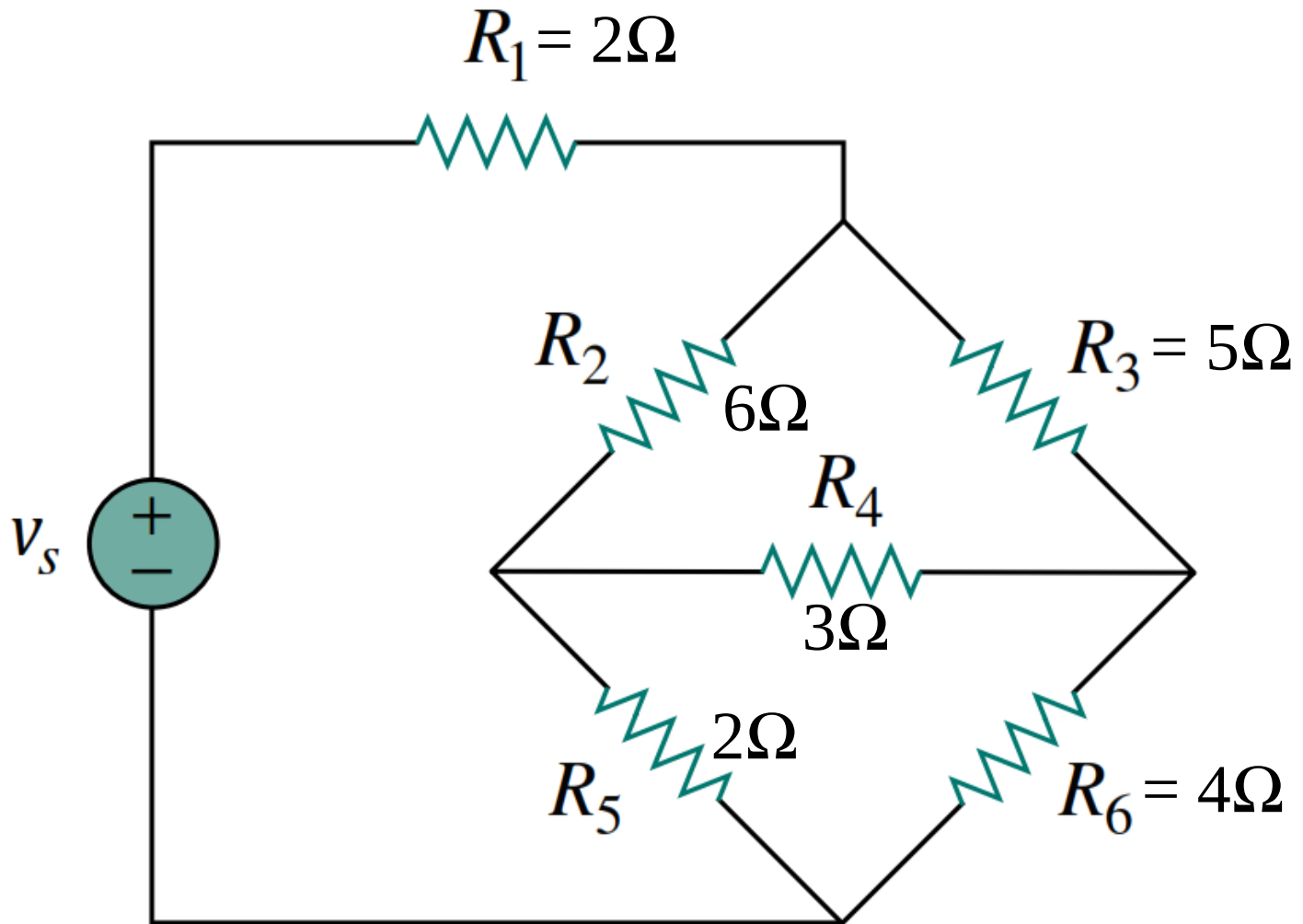


For the bridge network in Fig. 2.54, find R_{ab} and i .

Answer: $40\ \Omega$, 2.5 A .

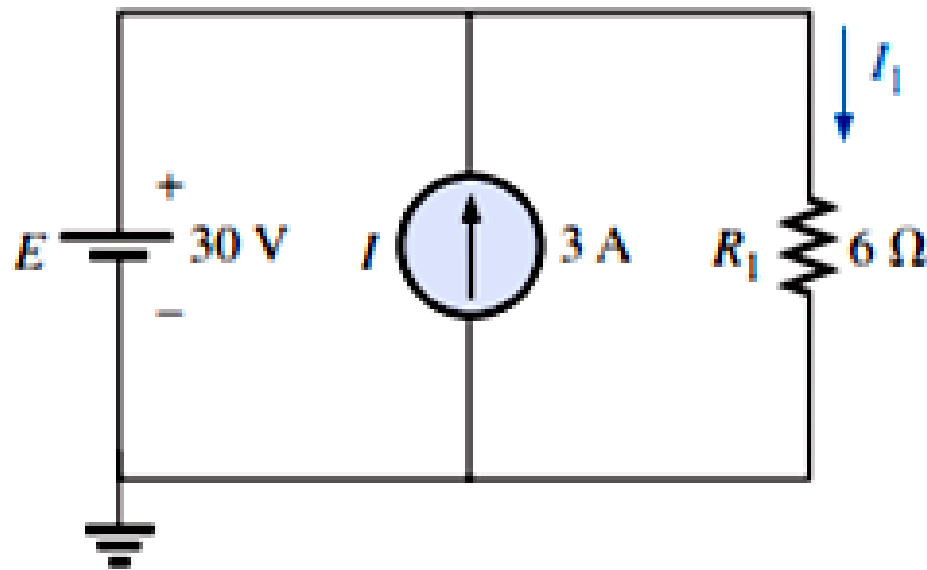


Find the total resistance.



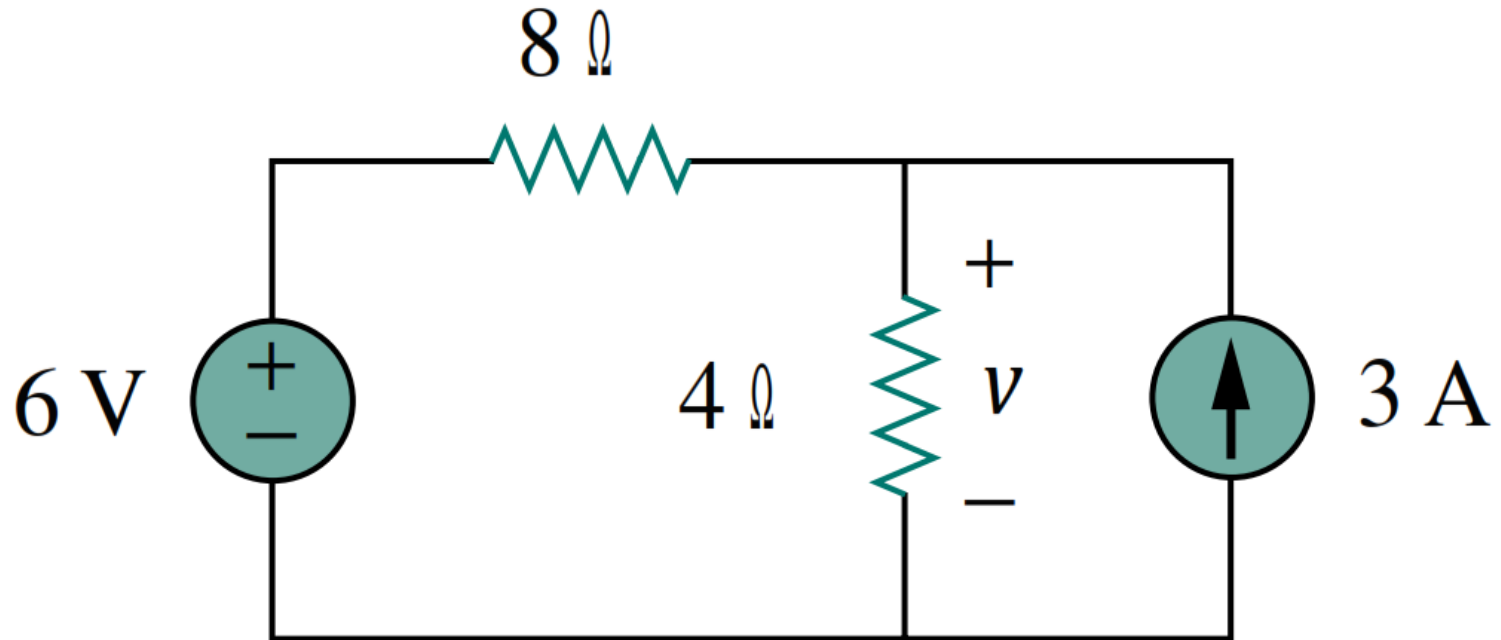
Superposition Principle

The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

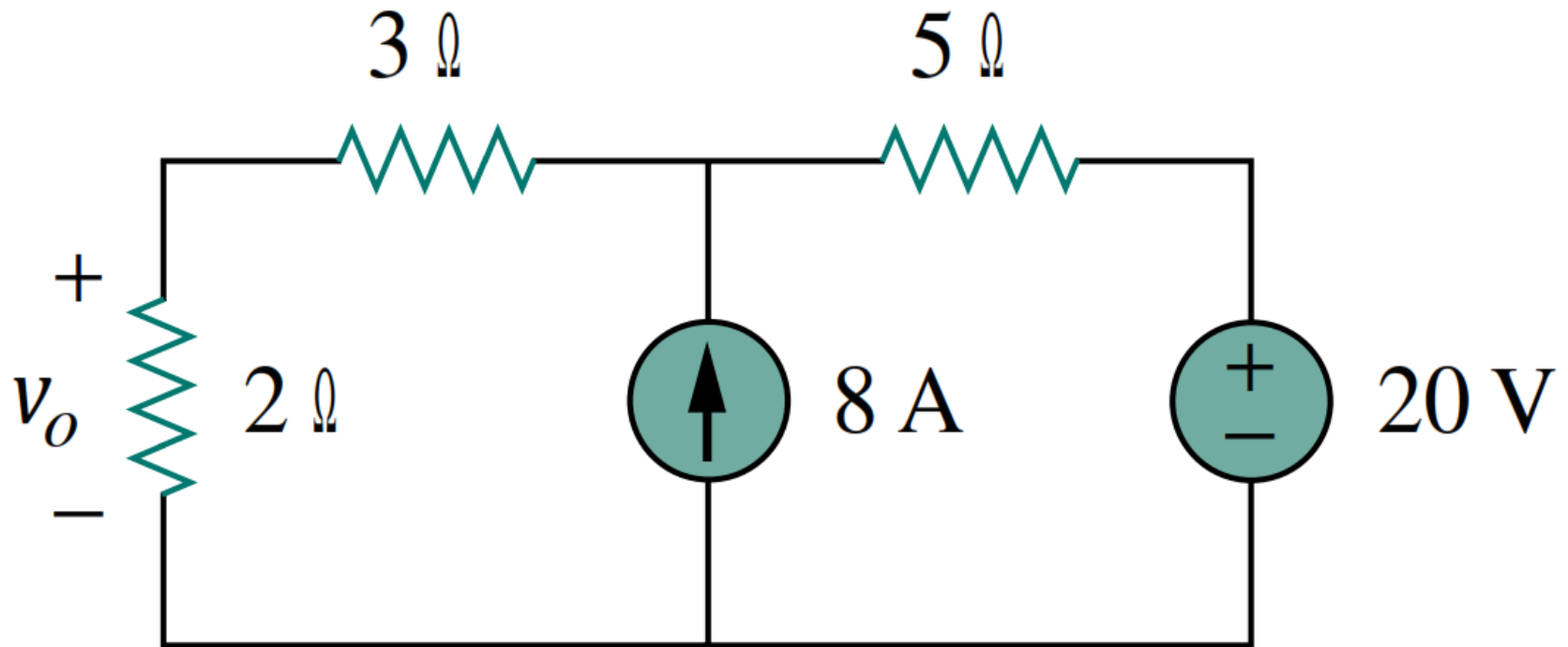


$$I_1 = I'_1 + I''_1$$

Use the superposition theorem to find v in the circuit



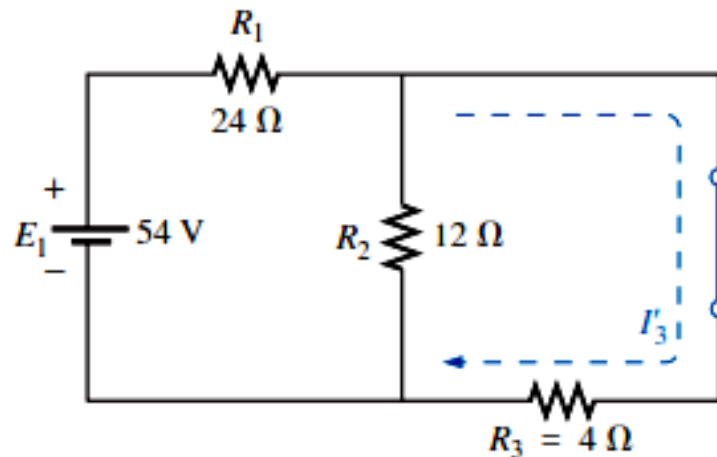
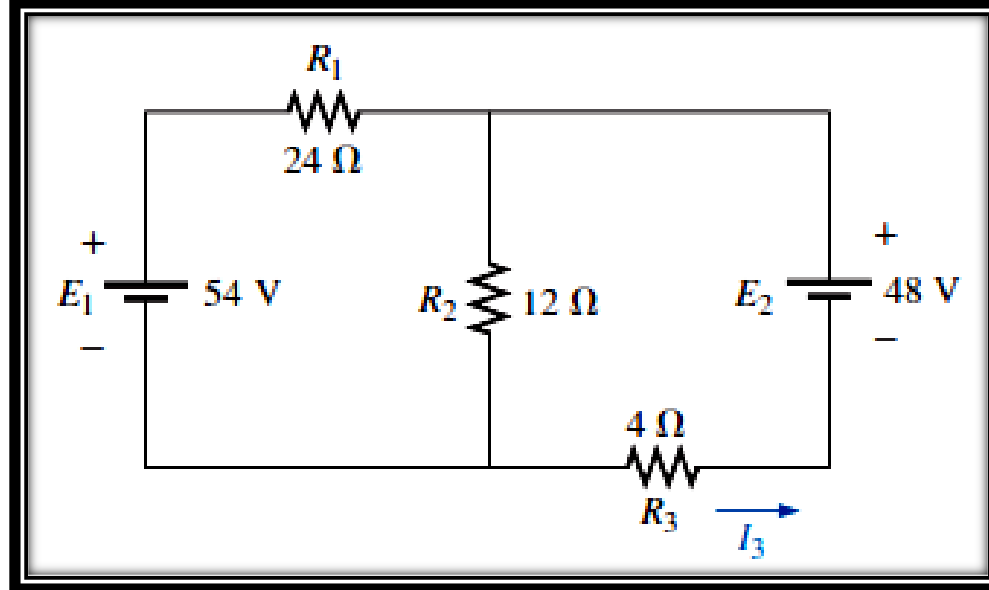
Use the superposition theorem to find v in the circuit



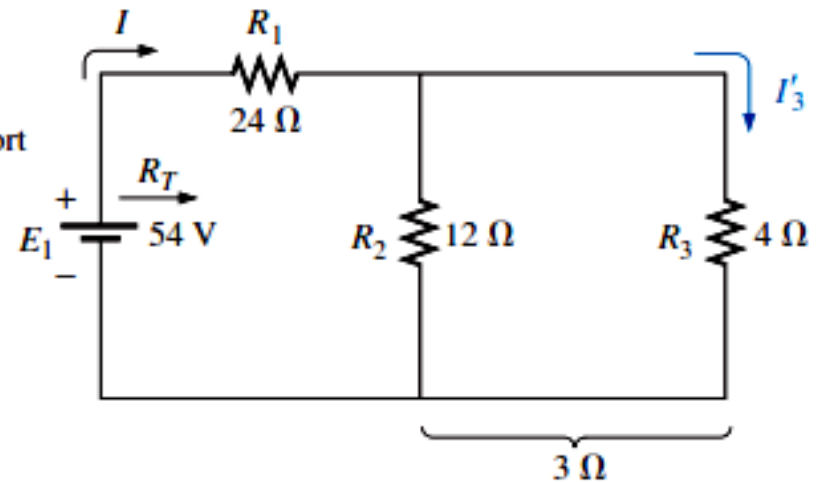


Problem

$$I_3 = ?$$



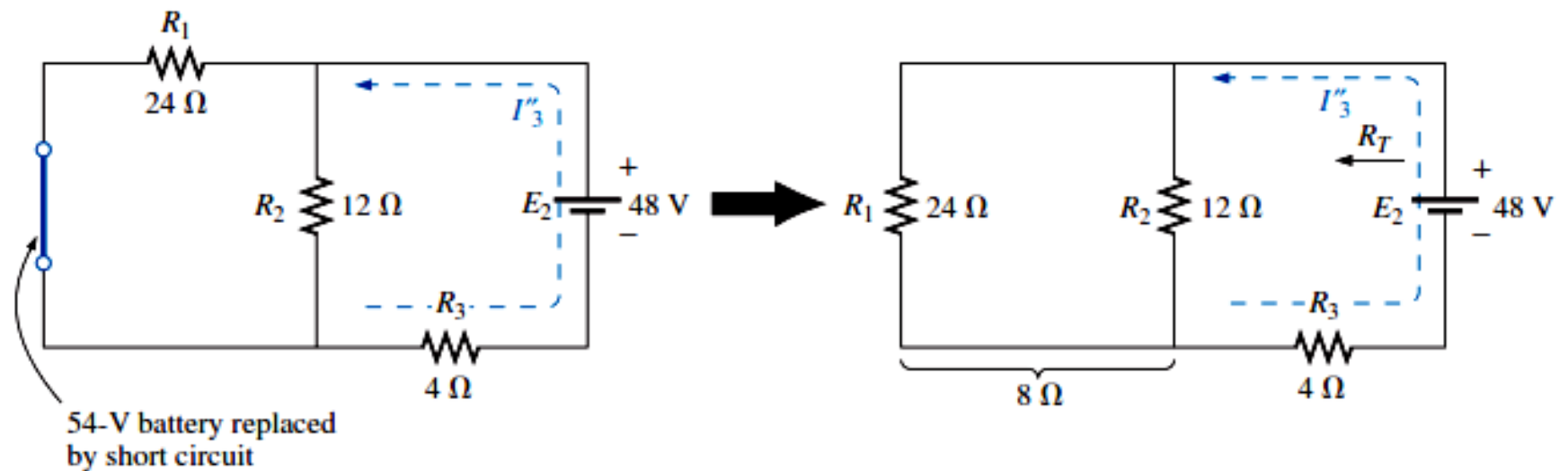
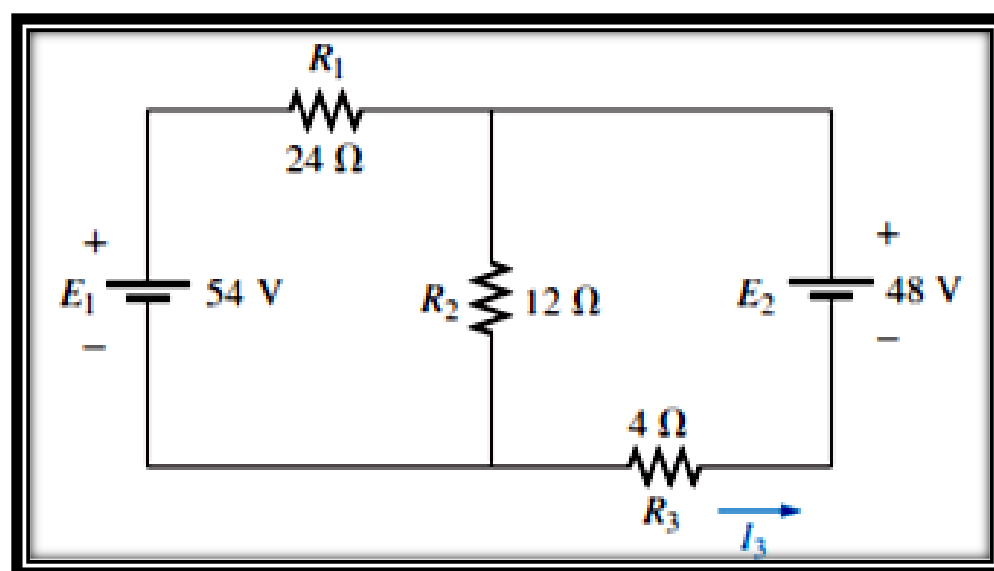
48-V battery
replaced by short
circuit



$$R_T = R_1 + R_2 \parallel R_3 = 24 \Omega + 12 \Omega \parallel 4 \Omega = 24 \Omega + 3 \Omega = 27 \Omega$$

$$I = \frac{E_1}{R_T} = \frac{54 \text{ V}}{27 \Omega} = 2 \text{ A}$$

$$I'_3 = \frac{R_2 I}{R_2 + R_3} = \frac{(12 \Omega)(2 \text{ A})}{12 \Omega + 4 \Omega} = \frac{24 \text{ A}}{16} = 1.5 \text{ A}$$



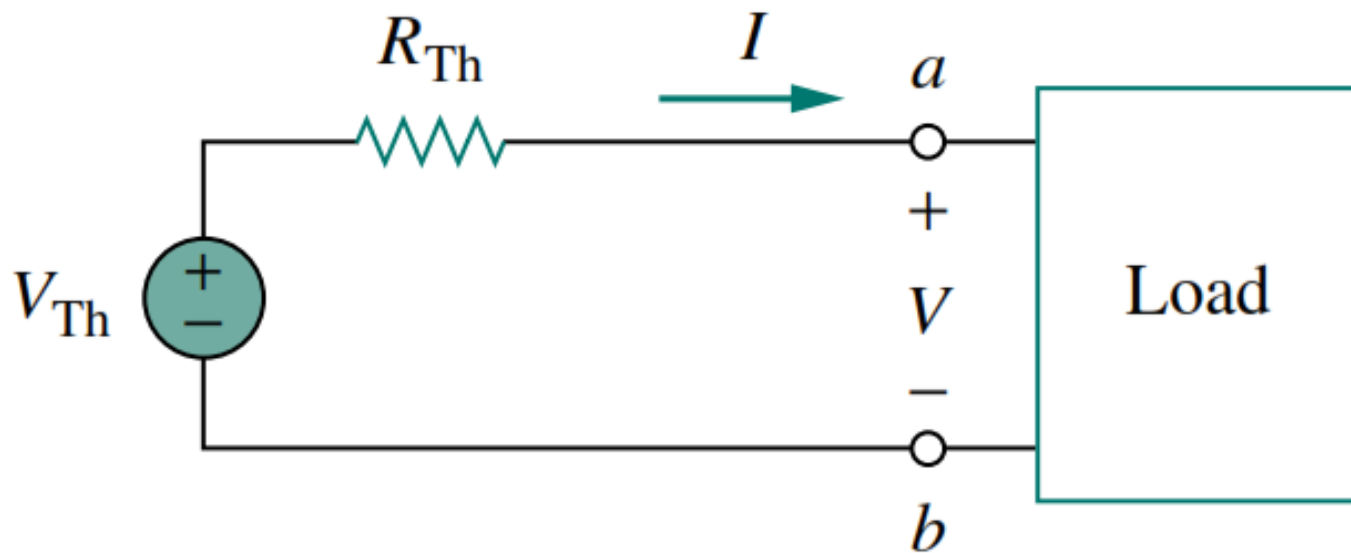
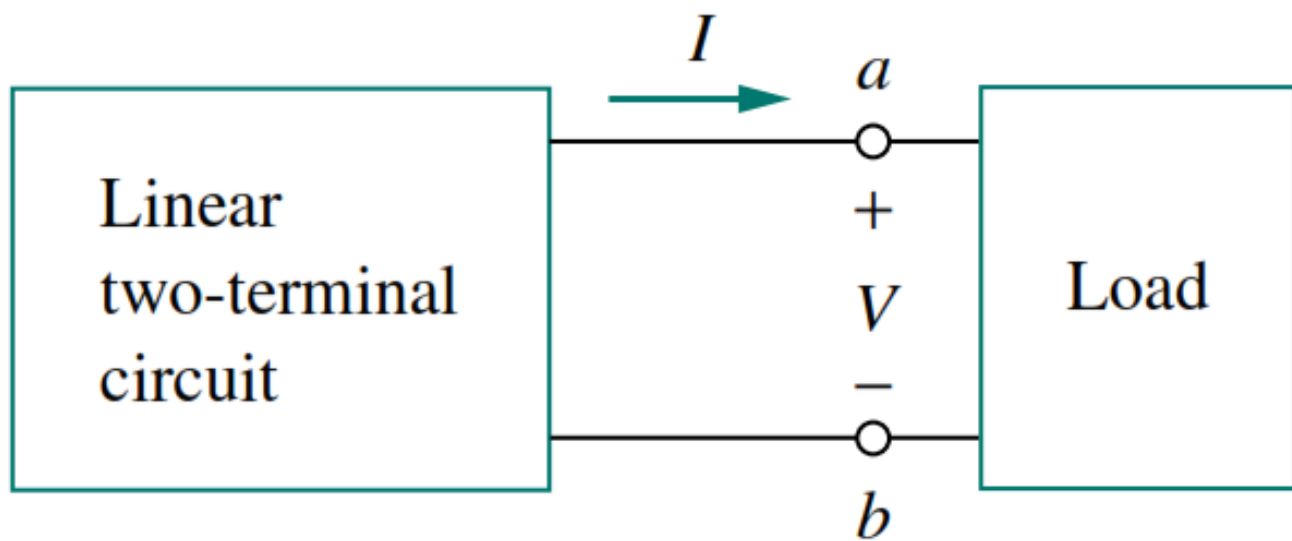
$$R_T = R_3 + R_1 \parallel R_2 = 4\ \Omega + 24\ \Omega \parallel 12\ \Omega = 4\ \Omega + 8\ \Omega = 12\ \Omega$$

$$I''_3 = \frac{E_2}{R_T} = \frac{48\ \text{V}}{12\ \Omega} = 4\ \text{A}$$

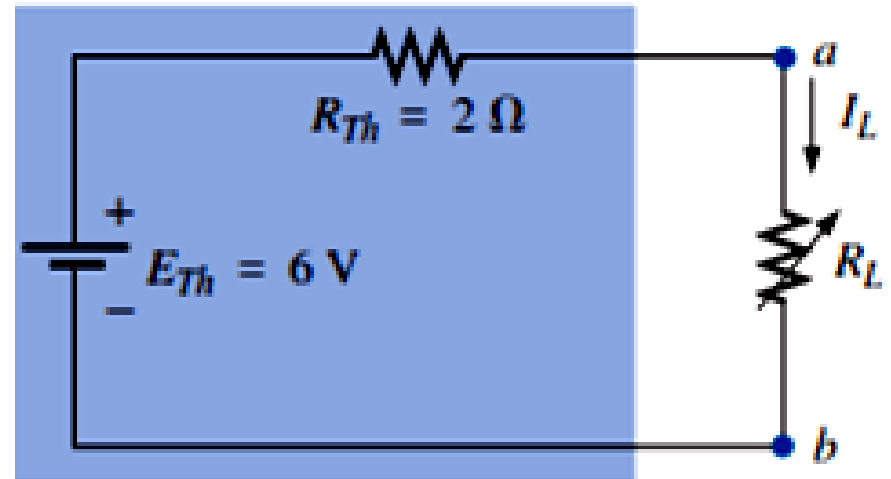
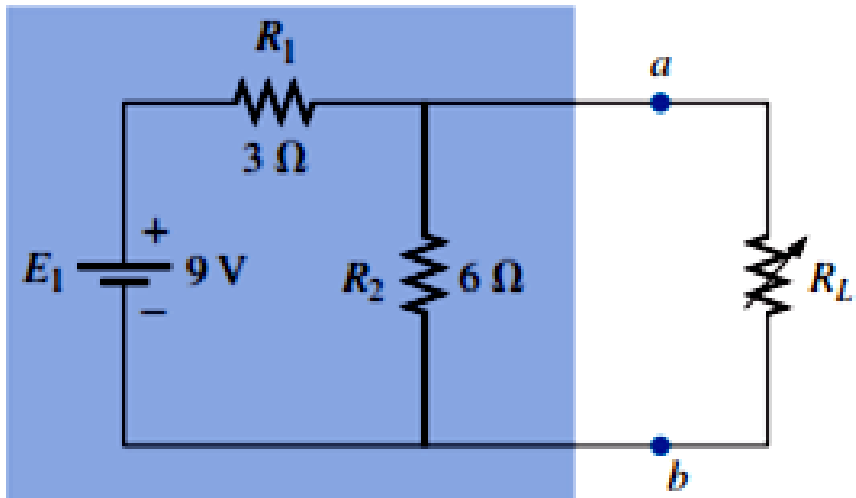
$$I_3 = I''_3 - I'_3 = 4\ \text{A} - 1.5\ \text{A} = 2.5\ \text{A}$$

THEVENIN'S THEOREM

Any two-terminal, linear bilateral dc network can be replaced by an equivalent circuit consisting of a **voltage source and a series resistor.**



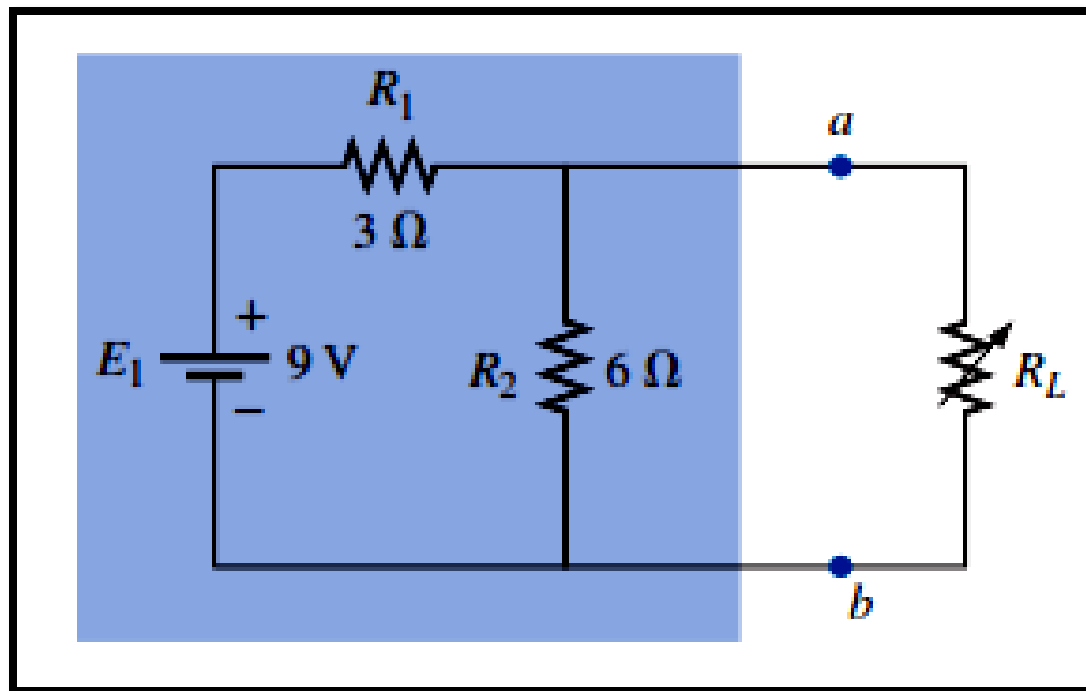
Example





Problem

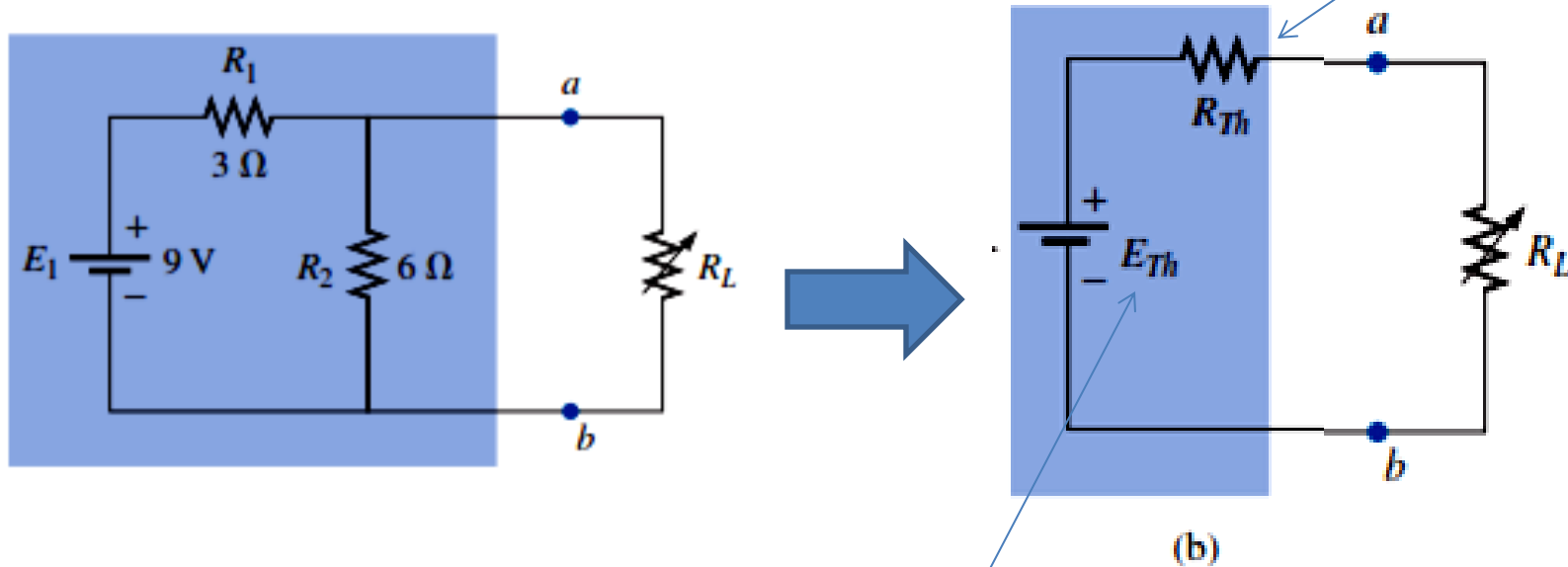
Find the Thevenin equivalent circuit



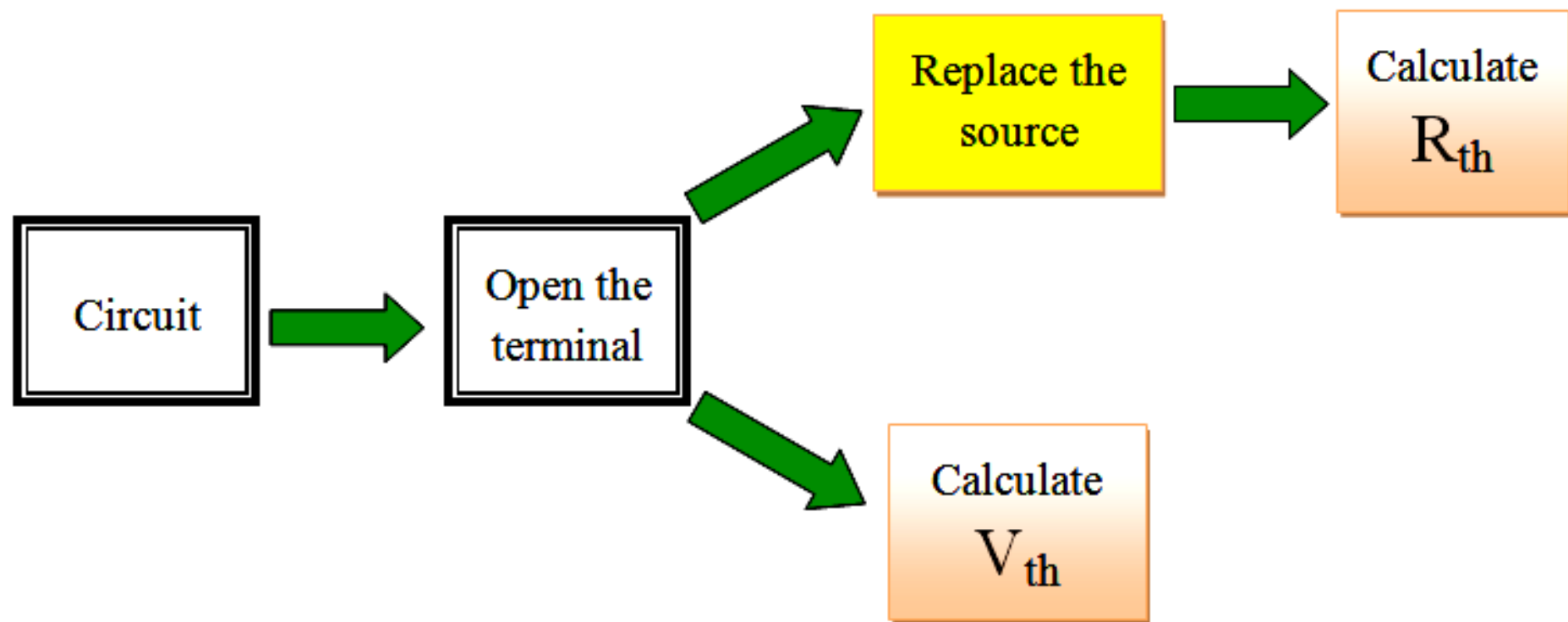
Solution:

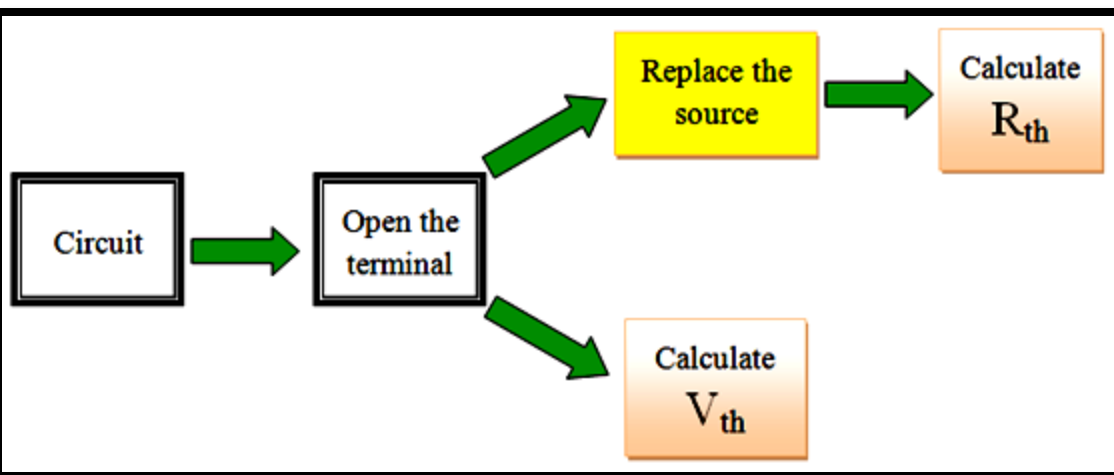
$$R_{th} = ?$$

The Thevenin equivalent circuit would be-

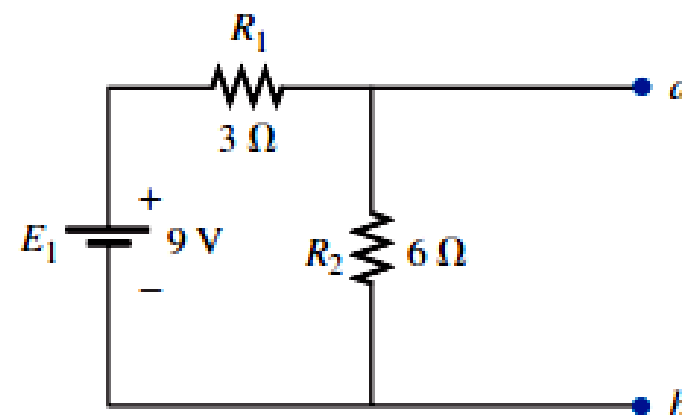
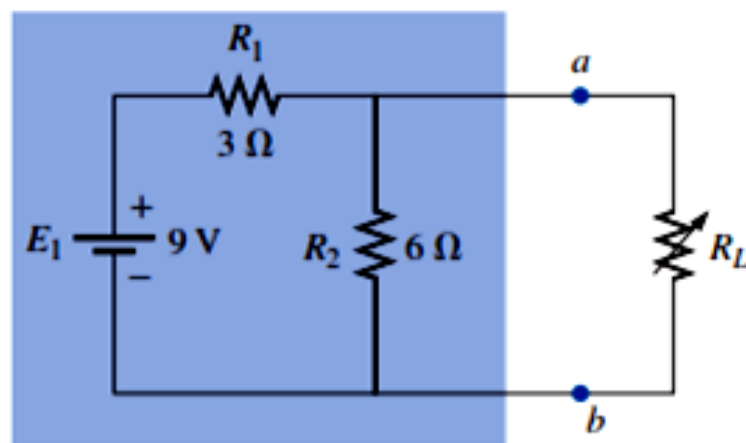


$$E_{th} = ?$$

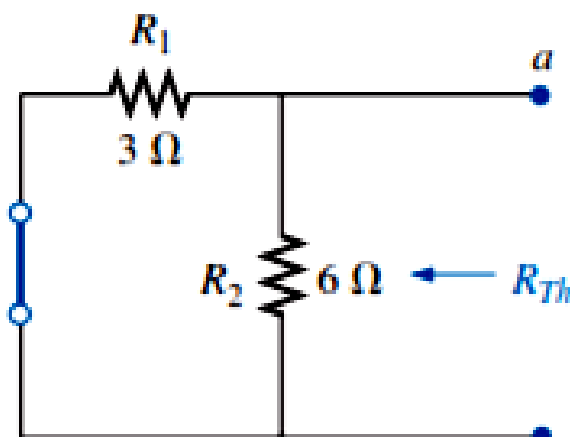




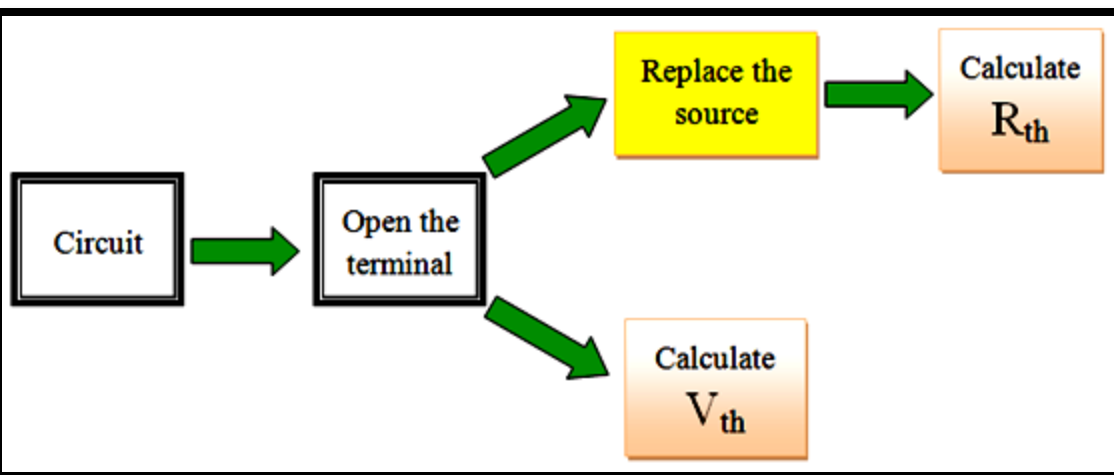
R_{th}



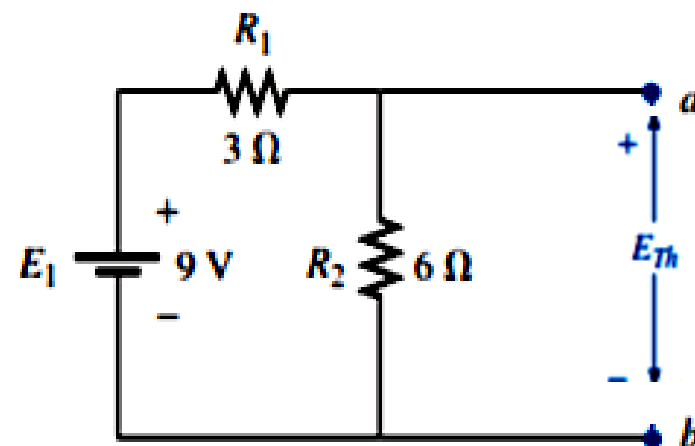
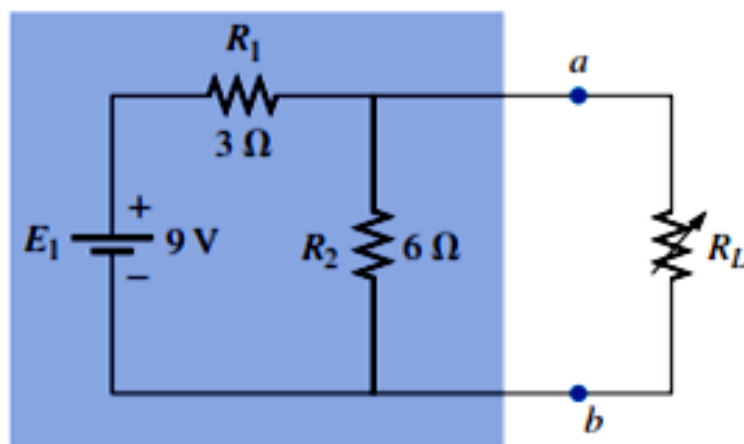
I source open
V source short



$$R_{Th} = R_1 \parallel R_2 = 2\Omega$$

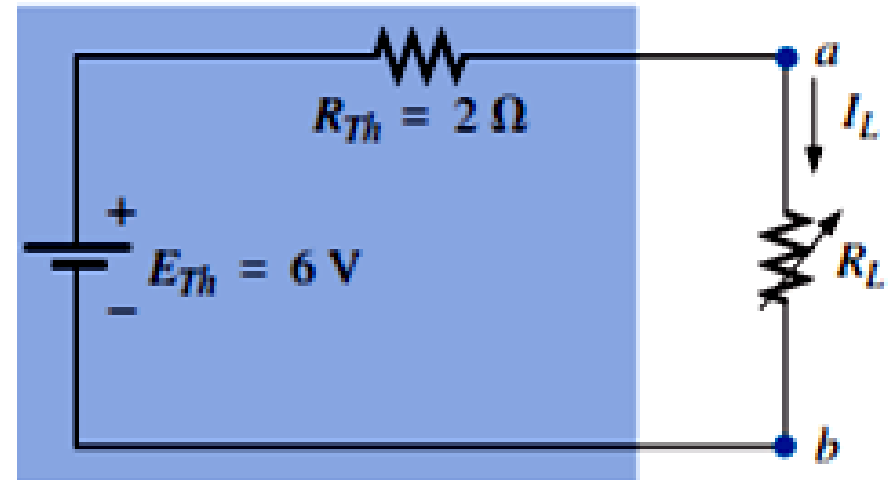
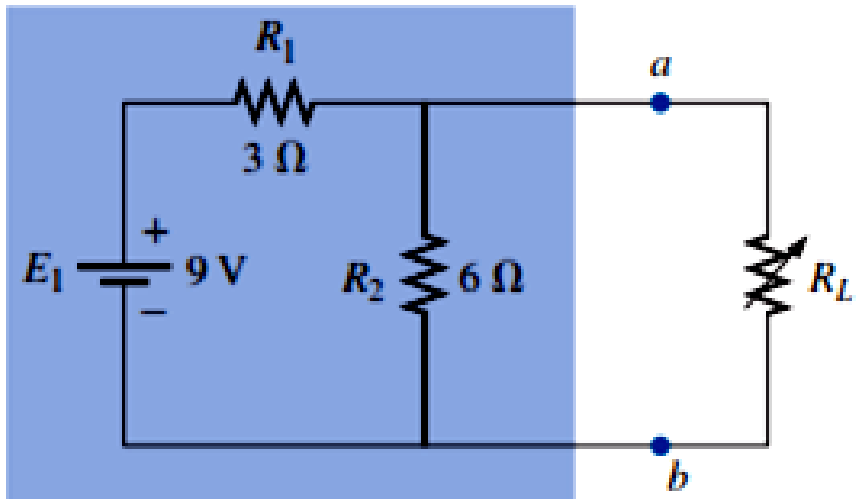


V_{th}



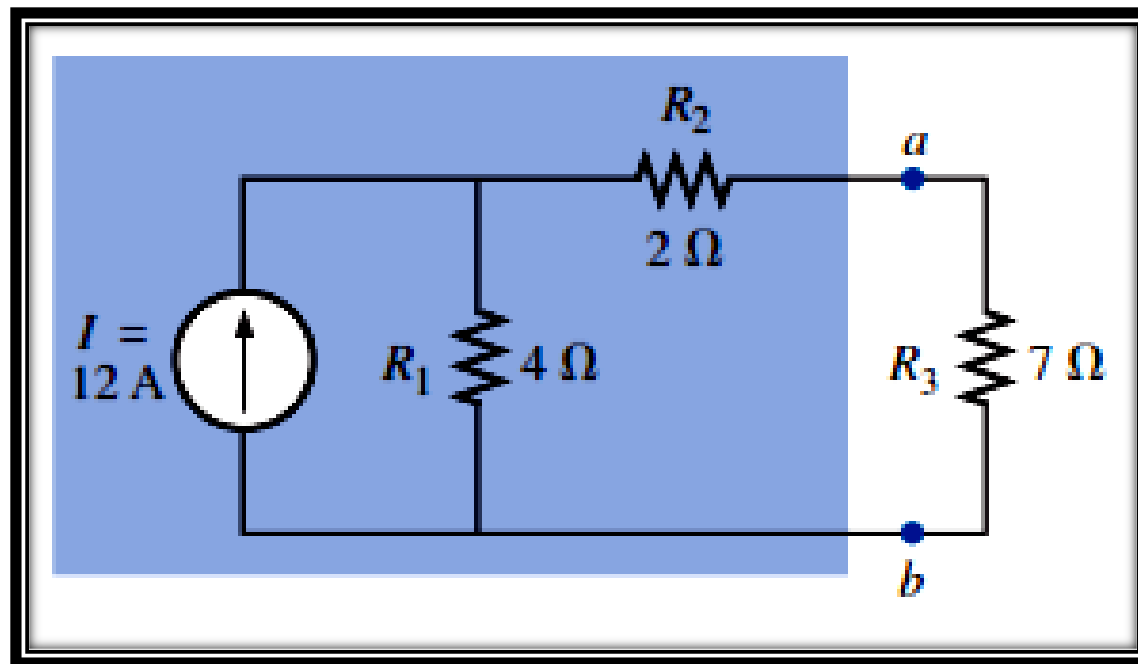
$$E_{Th} = \frac{R_2 E_1}{R_2 + R_1} = \frac{(6\ \Omega)(9\ \text{V})}{6\ \Omega + 3\ \Omega} = \frac{54\ \text{V}}{9} = 6\ \text{V}$$

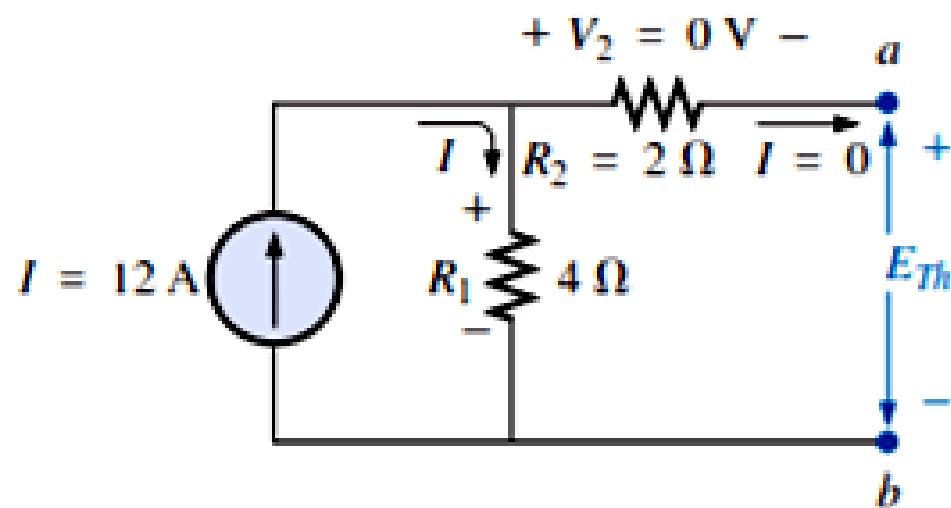
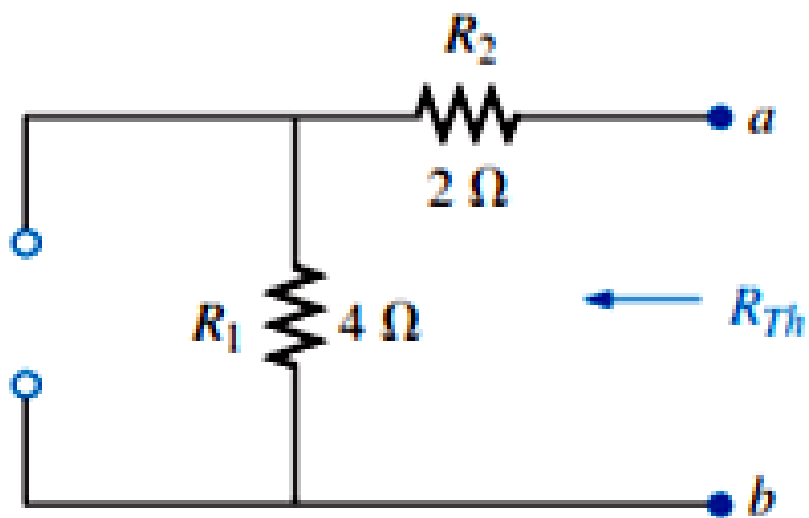
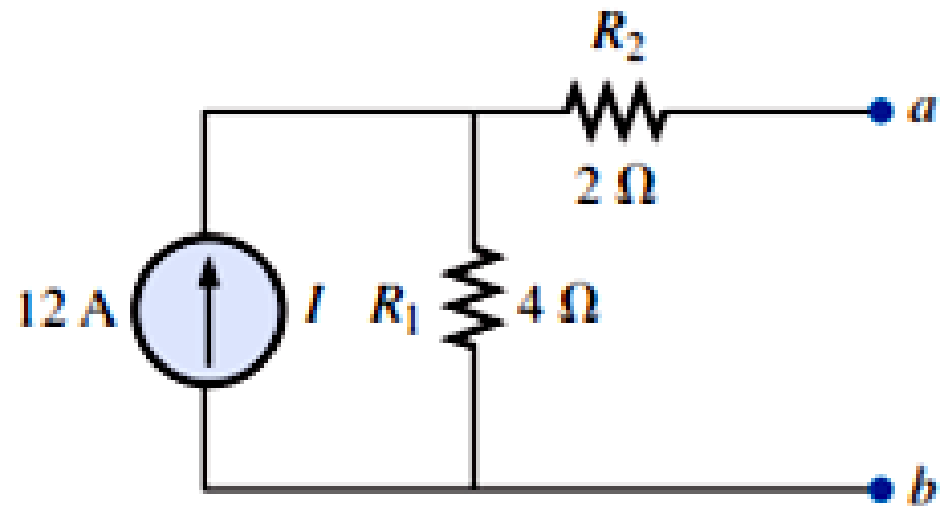
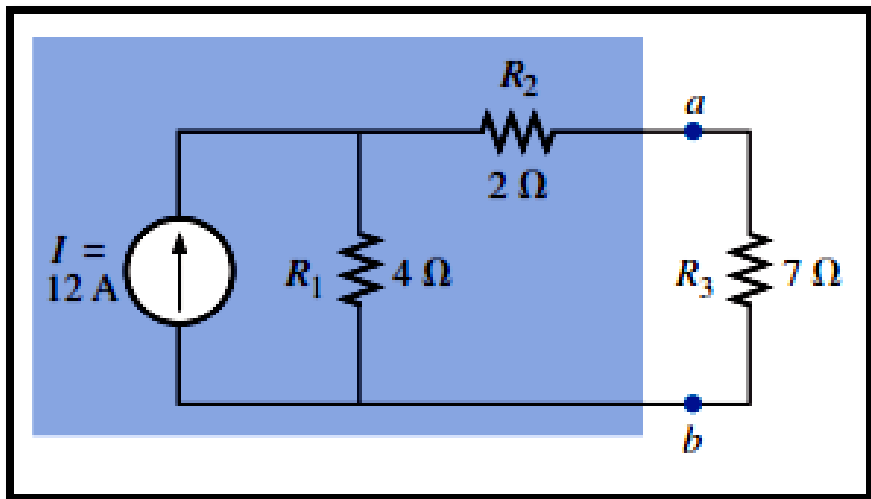
Answer



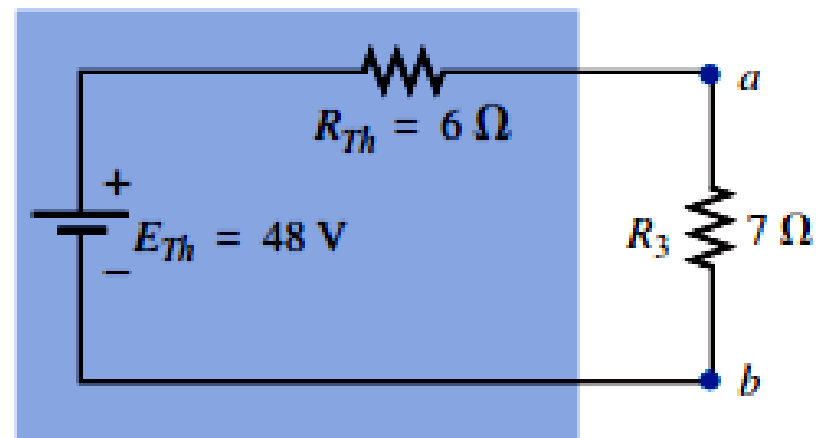
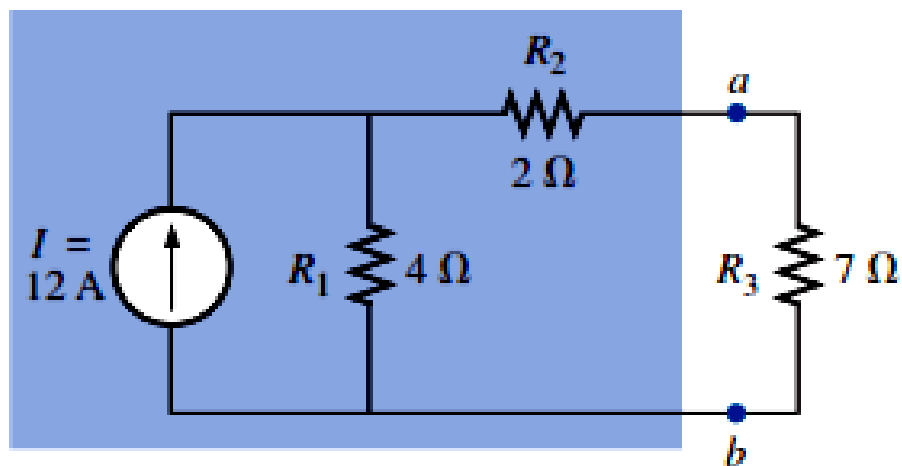
Problem ?

Find the Thevenin equivalent circuit



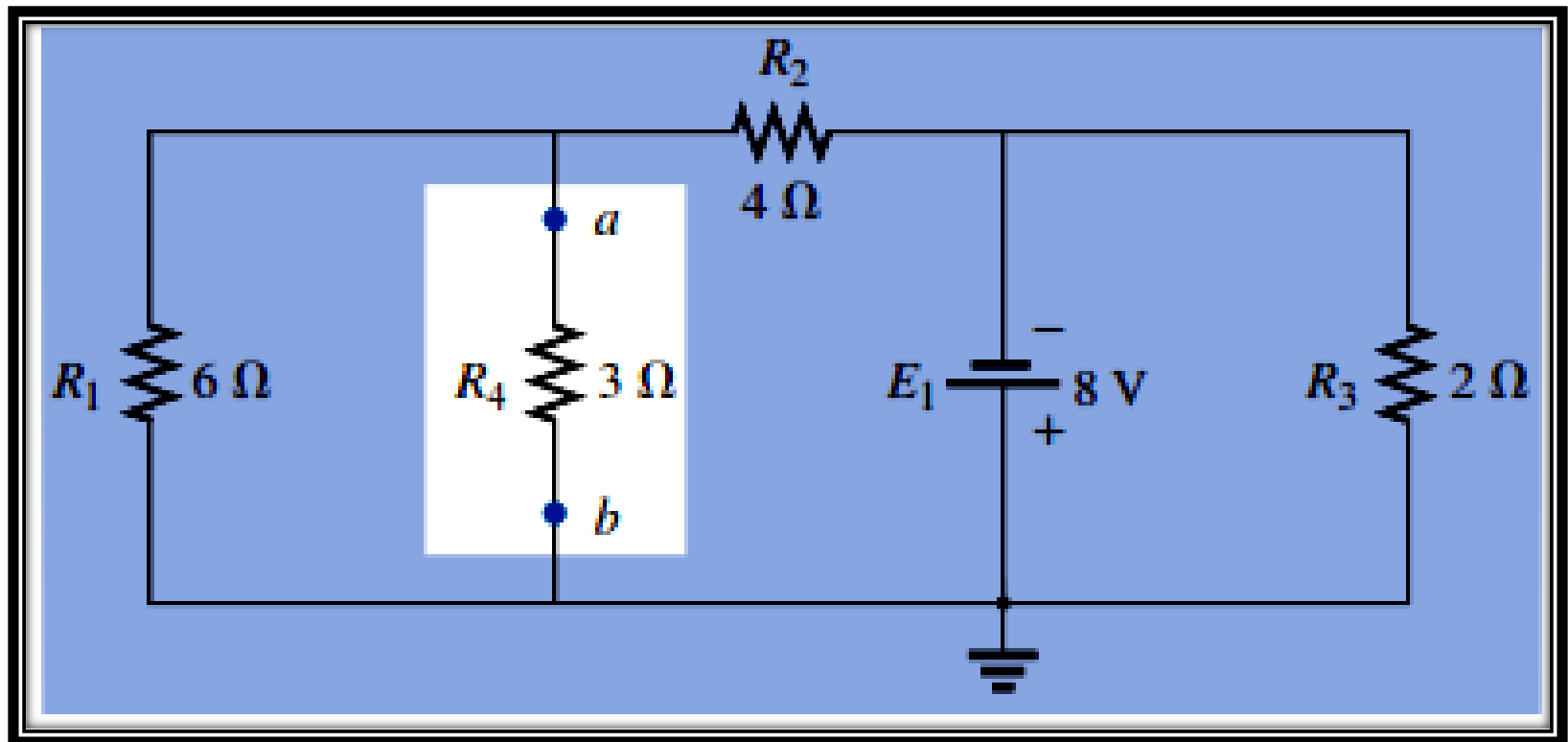


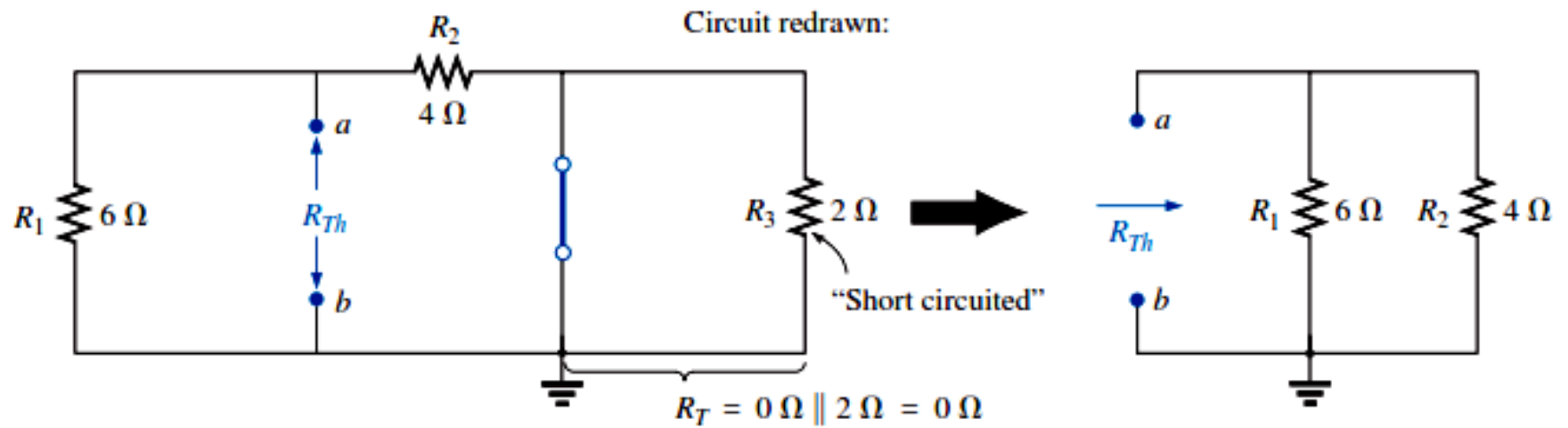
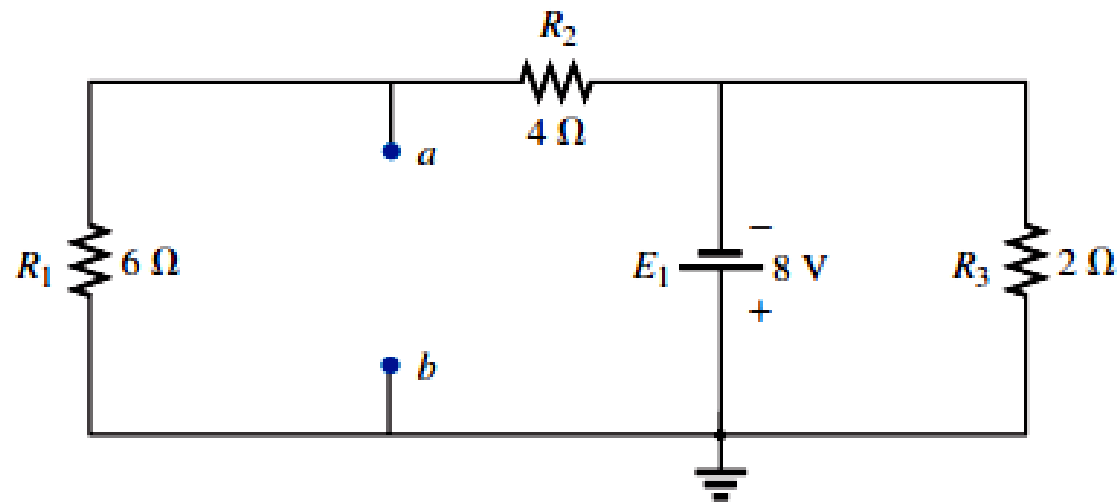
Answer

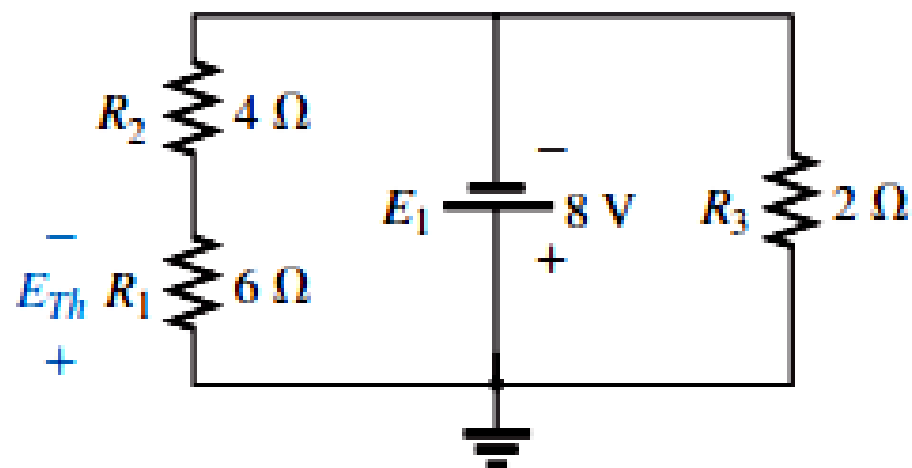
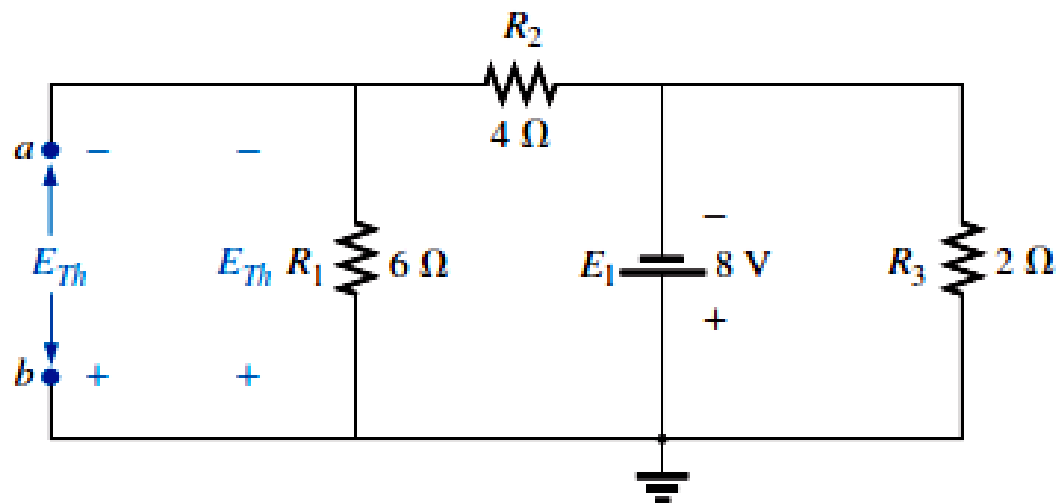
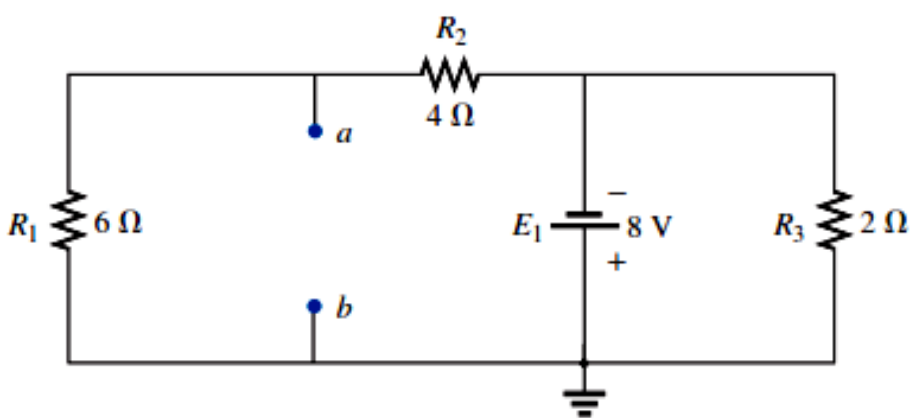


Problem ?

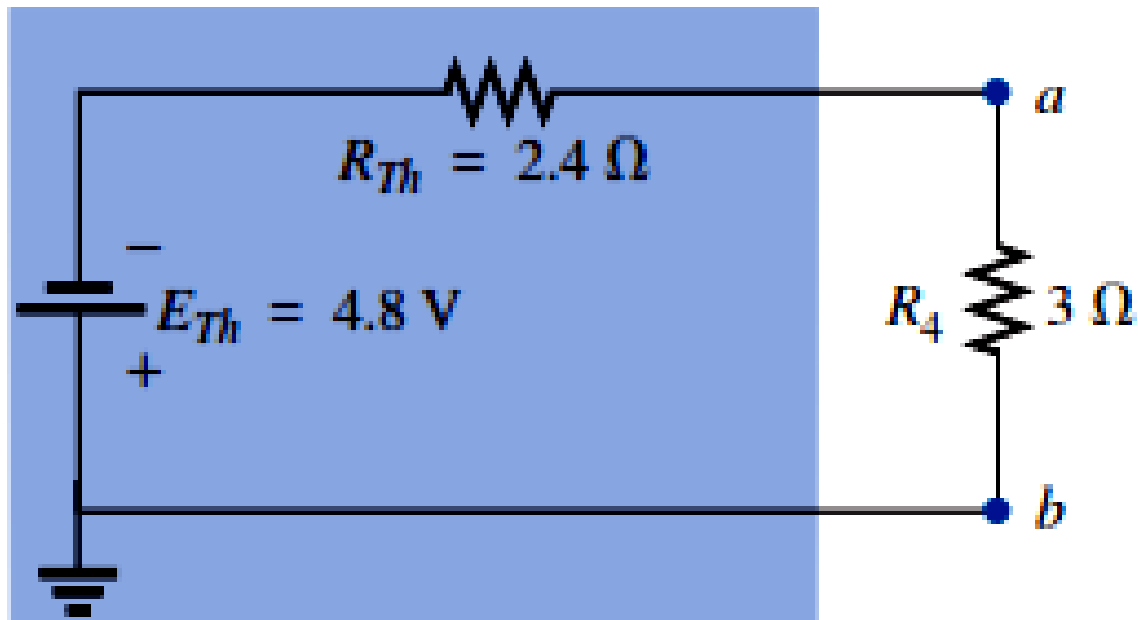
Find the Thevenin equivalent circuit





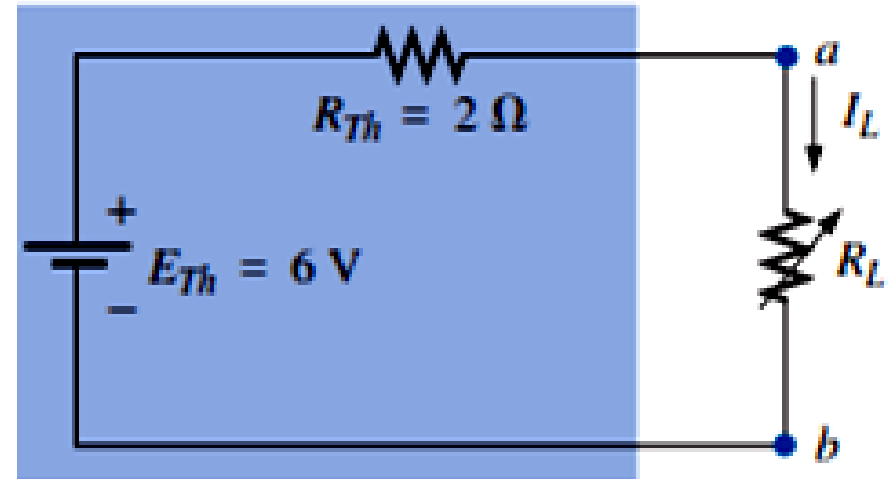
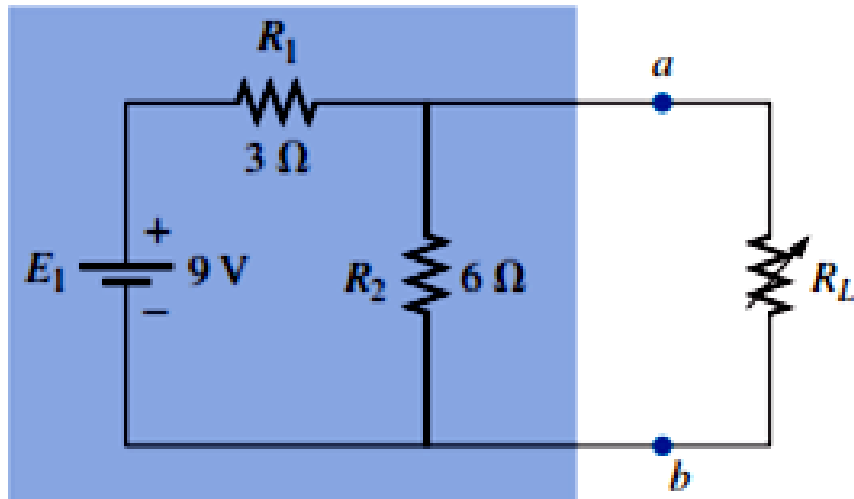


Answer



Maximum Power Transfer Theorem

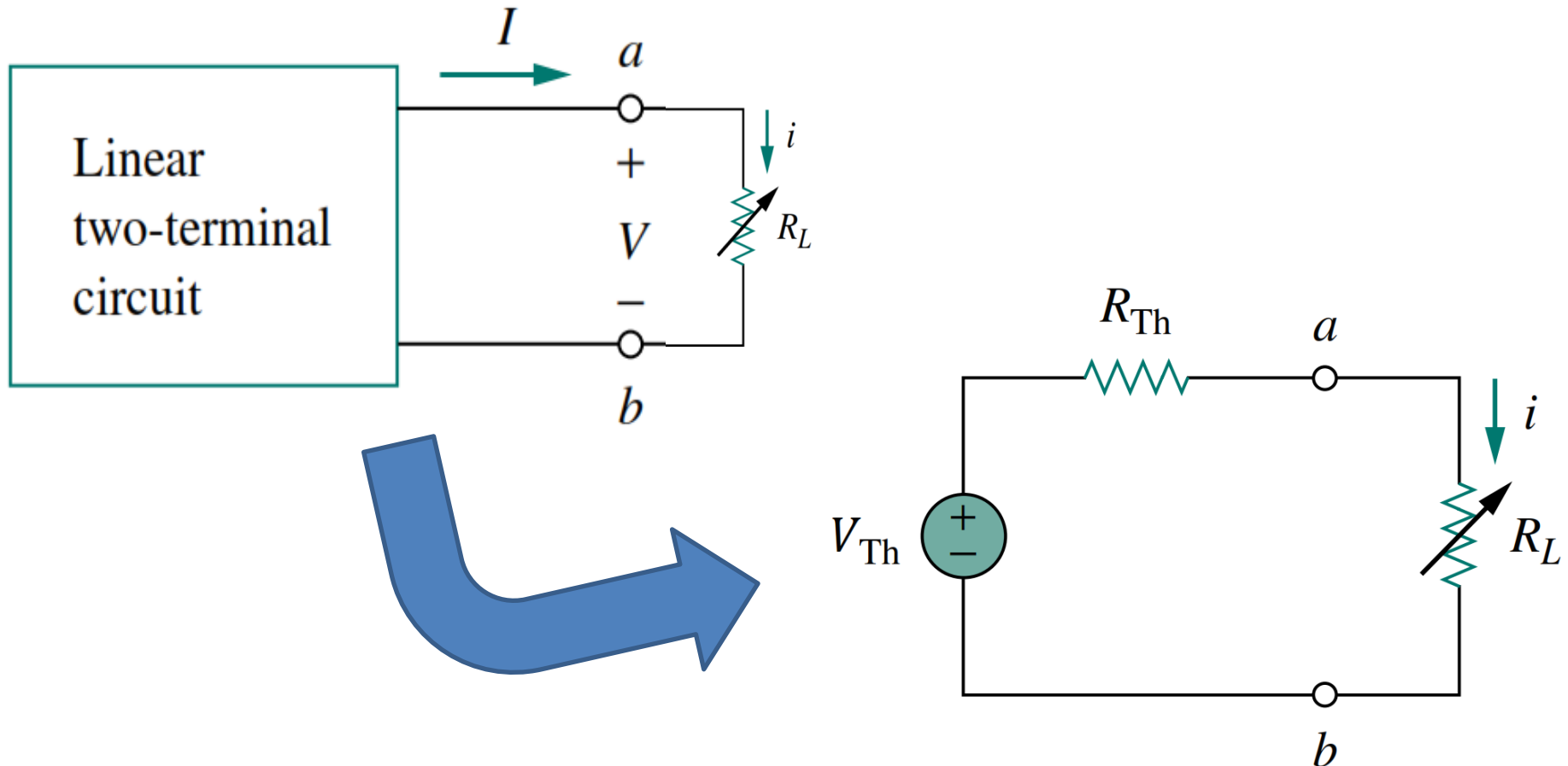
Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).



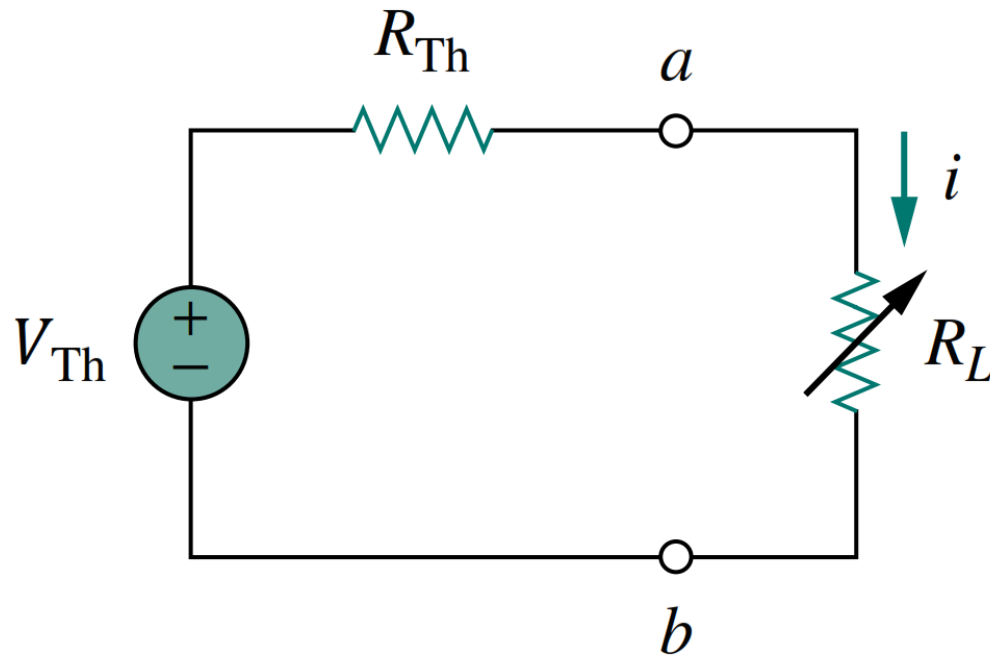
According to the theorem the load will receive maximum power when $R_L = R_{Th}$

Derivation of Maximum Power Transfer Theorem

First convert the circuit to its thevenin equivalent circuit



Now, the power delivered to the load is



$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

Now, the power (P) will be maximum when the differentiation of P with respect to R_L is zero.

$$\begin{aligned}\frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0\end{aligned}$$

This implies that

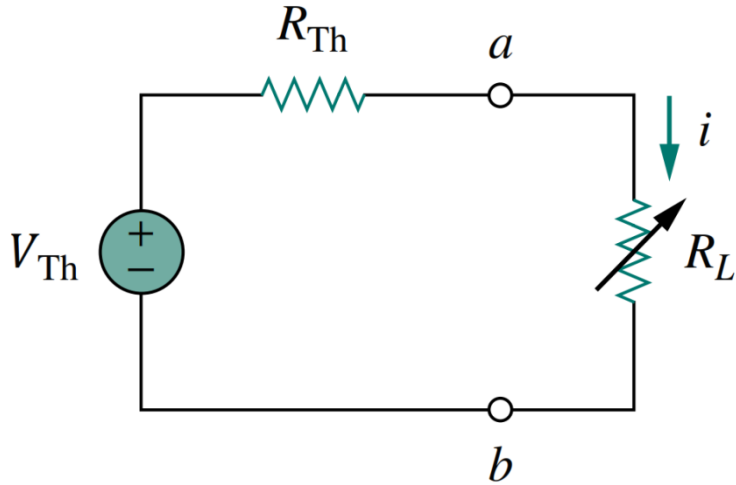
$$0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L)$$

which yields

$$R_L = R_{Th}$$

Proved

Now, the equation of maximum power-

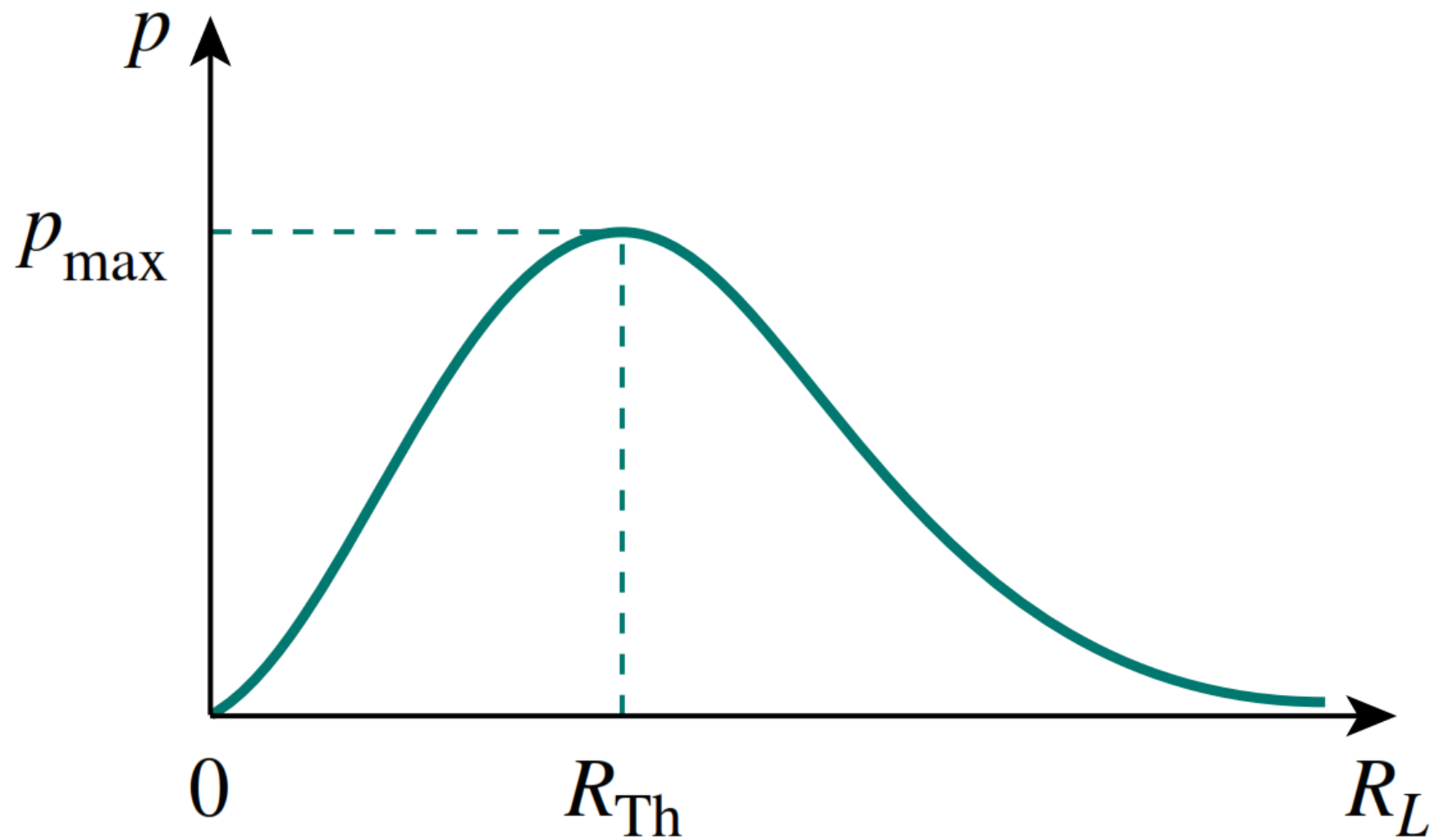


$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

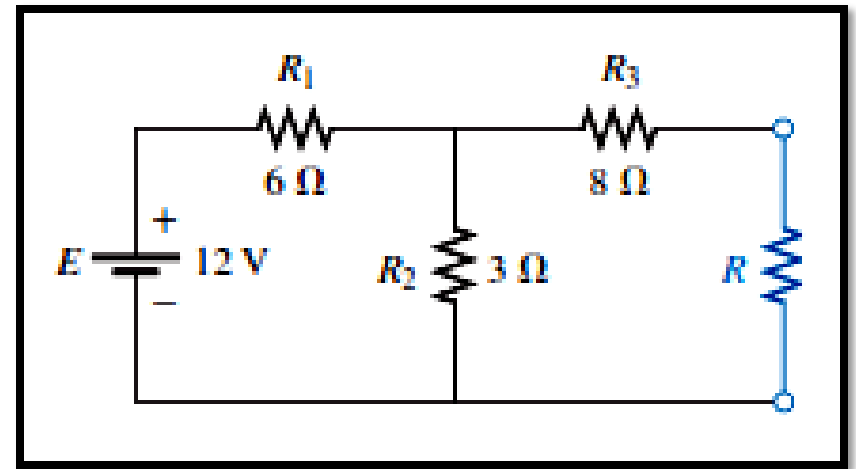
Putting $R_L = R_{Th}$ in the above equation we get,

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$

P vs R_L



Q. Determine the value of R for maximum power to R and calculate the power under this condition.



$$R_{Th} = R_3 + R_1 \parallel R_2 = 8\ \Omega + \frac{(6\ \Omega)(3\ \Omega)}{6\ \Omega + 3\ \Omega} = 8\ \Omega + 2\ \Omega$$

$$R = R_{Th} = 10\ \Omega$$

Ans

$$E_{Th} = \frac{R_2 E}{R_2 + R_1} = \frac{(3\ \Omega)(12\ \text{V})}{3\ \Omega + 6\ \Omega} = \frac{36\ \text{V}}{9} = 4\ \text{V}$$

$$P_{L_{\max}} = \frac{E_{Th}^2}{4R_{Th}} = \frac{(4\ \text{V})^2}{4(10\ \Omega)} = 0.4\ \text{W}$$

Ans

Problem



Thank You